

HARE-Rec: History-Aware Routing for Explainable Cold-Start Recommendation with Auditable Semantic-Atom Diagnostics

Xiuze Dong^a, Zhao Wang^{a,*}, Maosheng He^a, Wenxin Chen^b, Zhiqiang Liang^a and Kangfei Xiao^a

^aBeijing Electronic Science and Technology Institute, Beijing, 100070, China

^bShanghai University, Shanghai, 200444, China

ARTICLE INFO

Keywords:

Explainable recommendation

Cold-start

Neuro-symbolic learning

Knowledge-based systems

Auditability

ABSTRACT

Cold-start recommendation exposes a persistent tension between ranking accuracy and explanation faithfulness: collaborative models perform well for data-rich users but degrade under sparse histories, while knowledge-driven methods provide interpretable rationales yet often sacrifice ranking quality. Bridging this gap requires a model that adapts to heterogeneous user regimes while producing explanations tied to its scoring process.

We propose HARE-Rec, a regime-aware neuro-symbolic recommender with three functionally distinct components. A collaborative backbone (LightGCN) provides robust ranking signals, a semantic horn-rule reranker operates over text-derived item-level atoms to expose a knowledge-grounded audit interface, and a history-aware routing policy selects the scoring pathway at inference time based on each user's interaction history length, with a popularity fallback for super-cold users. All components are shared across regimes rather than duplicated for different user groups. We further assess auditability through counterfactual perturbations of explanation atoms.

Experiments on Amazon Video_Games 2023, Anime, and Yelp2018 with five random seeds show statistically significant gains over the neuro-symbolic baseline DiffFNN, including NDCG@10 improvements from 0.402 to 0.537 on Video_Games and from 0.180 to 0.631 on Yelp2018. On Video_Games, our method also outperforms recent strong baselines including ELCRec (NeurIPS 2024, 0.519) and AlphaRec (ICLR 2025, 0.451). Ablation studies show that routing primarily improves ranking robustness, while the semantic pathway supports auditability. On Yelp2018, removing the top-5 explanation atoms changes the final top-10 list for 41% of users and for 63% of high-contribution users.

Overall, HARE-Rec offers a single deployable architecture that adapts across cold and warm regimes while exposing auditable semantic traces for debugging and verification. The code and data artifacts are available at <https://github.com/wangzhao-cs/hare-rec>.

1. Introduction

Recommender systems now act as a critical decision layer in many online platforms, shaping what information, products, and services users are exposed to. As these systems enter increasingly diverse and regulated settings, two requirements become inseparable. The system must remain reliable when user histories are incomplete or rapidly changing, because a substantial fraction of real traffic originates from users with few interactions. At the same time, the system must provide explanations that stakeholders can understand and verify, because explanations are often the only practical interface for users, operators, and auditors to reason about model behavior.

Cold-start recommendation sits at the center of this tension. In warm regimes, collaborative filtering can exploit rich interaction signals and typically provides strong

ranking performance. However, as soon as histories become sparse, purely collaborative signals can collapse, pushing models toward popularity bias or unstable personalization. Knowledge-based and content-based methods alleviate this brittleness by grounding decisions in explicit attributes, rules, or semantics, which can also improve interpretability. Yet, in practice, the most interpretable rationale is not necessarily the most faithful one: a model can generate convincing post-hoc explanations that are only loosely coupled to its actual scoring mechanism. From the viewpoint of Knowledge-Based Systems, this disconnect is problematic because it undermines trust and makes it difficult to diagnose or improve system behavior.

This work is motivated by a practical question: can we build a recommender that remains robust under cold-start heterogeneity while producing explanations that are not only readable but also auditable with respect to the model mechanism? We do not seek to replace the strongest black-box recommenders in overall accuracy; rather, we ask whether a recommender can adapt to heterogeneous user regimes while exposing a mechanism-linked audit interface. We argue that achieving this goal requires treating user heterogeneity as a first-class systems problem rather than an afterthought. Instead of forcing a single scoring mechanism to cover all regimes, we combine complementary

*Corresponding author

Email: dongxz@besti.edu.cn (X. Dong); 20243821@mail.besti.edu.cn (Z. Wang); 20243809@mail.besti.edu.cn (M. He); wz0186@shu.edu.cn (W. Chen); 20242941@mail.besti.edu.cn (Z. Liang); 20253910@mail.besti.edu.cn (K. Xiao)

URL: <https://orcid.org/0009-0009-1131-069X> (Z. Wang)

mechanisms—collaborative representation learning and semantic rule-based reasoning—and govern them through an explicit regime policy determined by user history length. This yields a model that behaves like a strong collaborative recommender when data is sufficient, while retaining a semantic, knowledge-driven pathway for inspection, debugging, and auditing.

The central novelty is not merely the addition of a semantic reranker, but an auditable architectural decomposition with intentionally asymmetric roles: history-aware routing is responsible for robustness across user regimes, while the semantic pathway is responsible for exposing knowledge-level mechanism traces. Our empirical study supports the importance of this design. Across three benchmarks and five random seeds, our approach consistently improves over the neuro-symbolic baseline DiffFNN with statistically significant gains. On Amazon Video_Games 2023 (overall split), our full system (Ours3Stage) improves NDCG@10 from 0.402 ± 0.007 to 0.537 ± 0.003 . On Anime (overall split), we raise NDCG@10 from 0.493 ± 0.027 to 0.813 ± 0.003 . On Yelp2018 (overall split), we improve NDCG@10 from 0.180 ± 0.010 to 0.631 ± 0.006 . Crucially, on Video_Games our method also outperforms recent 2024–2025 baselines (ELCRec: 0.519, RLMRec: 0.460, AlphaRec: 0.451), demonstrating competitive accuracy alongside audibility. Importantly, these gains are primarily driven by the regime policy, especially the super-cold popularity fallback, whereas the semantic atoms and horn-rule reranker contribute mainly by making the decision process inspectable at the knowledge level. Beyond accuracy, we provide mechanism-linked diagnostics via semantic-atom perturbations, manual audit, and qualitative traces that expose which knowledge units and rules were used.

2. Related Work

2.1. Cold-start recommendation

Cold-start recommendation has been studied from multiple perspectives, including content-based and hybrid recommenders [1], meta-learning and transfer approaches [2], and models that exploit side information such as item attributes, text, and knowledge signals [3]. A recurring observation is that sparse histories reduce the reliability of purely interaction-driven personalization, and additional semantic evidence can stabilize ranking decisions. However, many cold-start methods focus primarily on accuracy and treat user histories as a discrete boundary (cold vs warm), even though real traffic exhibits a continuum of evidence levels. Implicit feedback methods [4] show that interaction frequency varies widely across users, and warm-up strategies bridge cold-to-warm transitions [5]. Surveys [6, 7] confirm that cold-start remains one of the most persistent challenges in deployment. Recent work has explored LLM-based solutions: ColdLLM [61] uses an LLM to simulate user behavior for cold-start items, while DisCo [60] applies disentangled contrastive learning across domains. Multimodal approaches

such as MoToRec [58] and M2VAE [59] leverage multi-modal tokenization and variational encoders to address item cold-start without relying on interaction histories.

2.2. Explainable recommendation

Explainable recommendation has developed along multiple lines, ranging from model-intrinsic explainability (where explanations are produced by the scoring model itself) to post-hoc rationalization and interface-oriented designs [8]. For knowledge-based applications, a central concern is whether an explanation is coupled to the decision mechanism rather than merely plausible. Faithfulness is often discussed in terms of input sensitivity or counterfactual behavior: when an explanation highlights a set of factors, perturbing those factors should tend to induce measurable changes in scores or rankings [9]. This perspective emphasizes verifiability and debugging value, which is particularly important in Knowledge-Based Systems. Post-hoc techniques such as LIME [10] and explicit factor models [11] approximate decision logic in a model-agnostic manner, while knowledge-graph methods [12] ground explanations in semantic entities and relations. A central desideratum is faithfulness: an explanation should reflect the model's actual reasoning rather than merely provide a plausible post-hoc story [9]. Faithfulness diagnostics commonly operationalize this through comprehensiveness (does removing highlighted features reduce the output?) and sufficiency (do highlighted features alone reproduce the output?). Alternative approaches include necessity and stability metrics, as well as counterfactual probing where individual inputs are perturbed to measure mechanistic sensitivity. EXP3RT [62] uses LLM-based personalized preference reasoning to produce faithful and accurate explanations grounded in review text. Our work adopts deletion and sufficiency probes over semantic atoms as a reproducible faithfulness diagnostic directly linked to the scoring mechanism. Importantly, we distinguish pathway-level sensitivity (whether highlighted atoms materially affect the semantic score inside the reranker) from ranking-level sensitivity (whether those perturbations are large enough to alter the final ranking after collaborative and routed components are taken into account). Ranking-level effect depends on how much the semantic term contributes to the final score, which varies across datasets and user regimes.

2.3. Neuro-symbolic and rule-based recommendation

Neuro-symbolic recommendation seeks to reconcile representation learning with explicit reasoning, for example through logic constraints, rule induction, differentiable inference, or knowledge-graph reasoning [13, 14]. DiffFNN [15] introduced differentiable fuzzy Horn rules for recommendation, enabling end-to-end learning of interpretable rule structures. Such approaches align naturally with knowledge-centric deployment requirements because they can expose symbolic structures (atoms, rules, paths) that are easier to inspect than opaque embeddings alone. At the same time,

symbolic components can suffer under extreme sparsity when the relevant structures are weakly supported by data, and they may lag behind strong collaborative baselines if they are used as the sole scoring pathway. Propagation-based approaches such as RippleNet [16] diffuse preferences through knowledge graph ripples, while unified frameworks [17] optimize graph and collaborative signals jointly. Path-based reasoning methods [18] use reinforcement learning to generate interpretable recommendation paths over knowledge graphs, offering strong explainability but requiring entity-aligned knowledge graphs that may not be available in all domains. Knowledge-aware graph neural networks [19] enrich collaborative signals with structured relational features. Our work differs from these in that semantic atoms are extracted from item text rather than from a pre-existing structured knowledge graph, making the approach applicable when entity-aligned KGs are unavailable.

2.4. Positioning of this work

Our work is positioned at the intersection of cold-start recommendation, mechanism-linked explanation diagnostics, and neuro-symbolic reranking. We are not proposing a direct substitute for structured knowledge-graph reasoning systems; rather, the symbolic layer is built over text-derived semantic atoms and serves as a no-KG semantic audit interface. At the same time, we do not position the method as a pure black-box accuracy competitor. Instead, the contribution is a regime-aware, routing-enabled recommender that adapts to heterogeneous user regimes while exposing an auditable semantic pathway. This positioning distinguishes our work from both post-hoc explainability methods and knowledge-graph recommenders: our emphasis is on knowledge-grounded auditing of recommendation decisions, not on claiming that semantic reasoning is the sole or dominant source of ranking gain.

Our history-aware routing shares a surface resemblance with Mixture-of-Experts (MoE) architectures [47, 48], which learn a gating network to blend multiple expert sub-models. However, the two designs differ in three key respects. First, MoE systems learn a parameterized gating function end-to-end, whereas our routing is a fixed, rule-based policy determined entirely by the user’s history length—no additional parameters are introduced and no gating network is trained. This makes the routing decision fully transparent and auditable: for any user, the active scoring pathway can be read off from the history count alone. Second, MoE experts are typically homogeneous (e.g., identical architectures with different weights), while our routing selects among *heterogeneous* components—a popularity baseline, a neuro-symbolic DiffFNN scorer, and a collaborative-plus-semantic Two-Stage model—each designed for a different evidence regime. Third, our routing is motivated by the cold-start continuum rather than by capacity scaling: its purpose is to match the scoring pathway to the available evidence level, not to increase model capacity. We empirically validate that this simple heuristic is robust across threshold settings (see

the cold-start sensitivity analysis in Table 5) and competitive with a stronger heuristic routing baseline (Table 4).

2.5. Collaborative filtering and deep recommendation

Collaborative filtering [20, 21] and matrix factorization [22, 23] underpin most modern recommender systems. Efficient implicit feedback methods [4, 24] and sparse linear models [25] scale these ideas to large settings. Deep learning has advanced recommendation through variational autoencoders [26], graph convolutional networks [27], hybrid architectures [28, 29], and sequential models [30, 31]. Comprehensive surveys [32, 33] cover the breadth of these developments. Despite strong accuracy, most collaborative methods offer limited interpretability. Independent audits of neural recommendation approaches [34] highlight the importance of rigorous evaluation protocols [35, 36].

2.6. LLM-enhanced recommendation

Large language models (LLMs) have recently been integrated into recommender systems in several paradigms [52]. *LLM-as-feature* methods use LLM embeddings or generated text as auxiliary item/user features: RLMRec [43] aligns LLM-derived representations with collaborative signals, AlphaRec [50] injects LLM item embeddings into a contrastive LightGCN backbone, and LLMRec [53] augments interaction graphs with LLM-generated knowledge. *LLM-as-scorer* methods fine-tune LLMs directly for ranking: TALLRec [54] aligns language models for pairwise preference prediction, while CALRec [55] uses contrastive alignment for sequential recommendation. *LLM-as-knowledge* approaches [56, 57] leverage LLMs to bridge open-world knowledge with collaborative patterns. ELCRec [49] combines SASRec-style sequential modeling with LLM-enhanced contrastive learning for cold-start users. Our work differs from these approaches in that we do not use LLMs for scoring or retrieval; instead, we use LLM-extracted semantic atoms as a structured knowledge layer that enables auditable, mechanism-linked diagnostics. This distinction is important: while LLM-enhanced methods achieve strong accuracy, they do not inherently expose an interpretable reasoning pathway, whereas our semantic horn-rule reranker provides a compact, traceable audit interface grounded in discrete atoms and rules.

3. Problem Setting

We consider implicit recommendation over a set of users and items. Given a user, the task is to rank candidate items such that the observed future interaction items are ranked higher than unobserved items. We evaluate ranking quality with standard top- K metrics (HR, Recall, NDCG) under a unified protocol shared across all methods to ensure fair comparison.

To study cold-start systematically, we partition evaluation users by their history length. To avoid any potential leakage, the history length is computed using only interactions observed before the test window (i.e., from the training

period). We define cold users as those whose pre-test history length falls within the bottom quantile of this training-history distribution. In our experiments, we use a quantile of 0.2, which yields a dataset-adaptive cold threshold and enables consistent reporting across datasets.

This quantile-based protocol results in markedly different user-regime compositions across benchmarks, which motivates history-aware routing as a system-level policy. Using the dataset-adaptive thresholds (Video_Games: $m = 3$, Anime: $m = 55$, Yelp2018: $m = 8$), we observe that Video_Games contains a substantial super-cold population (18.7% of evaluation users have $|\mathcal{H}_u| \leq 2$), while Anime and Yelp2018 contain much smaller super-cold populations (Anime: 1.2%, Yelp2018: 2.8% with $|\mathcal{H}_u| = 0$). Under the same definition, the remaining users are split into cold (Video_Games: 13.4%, Anime: 19.3%, Yelp2018: 21.2%) and warm (Video_Games: 68.0%, Anime: 79.5%, Yelp2018: 76.0%). The cold/warm split reported in Table 2 groups super-cold and cold together (Video_Games: 32.1% cold side, 67.9% warm side). Our goal is not a single universal scoring rule, but a routing-enabled recommender that adapts to heterogeneous evidence levels while remaining inspectable through mechanism-linked semantic atoms.

4. Method

Our framework couples collaborative learning with semantic reasoning in a two-stage architecture, and it introduces history-aware routing to adapt the scoring pathway to each user's regime (Figure 1). We denote users by $u \in \mathcal{U}$ and items by $i \in \mathcal{I}$. Each user has an interaction history $\mathcal{H}_u \subseteq \mathcal{I}$, and the task is to rank candidate items for recommendation. The final score for user u and item i is computed by routing between complementary scoring functions: a collaborative backbone, a semantic horn-rule term injected through a Two-Stage reranker with a learned history-aware gate, and a cold-start routing layer that blends pathway scores based on the user's history length $|\mathcal{H}_u|$.

4.1. Collaborative backbone

The collaborative stage learns latent representations for users and items from the observed interaction graph. This backbone captures strong co-occurrence patterns and provides a robust scoring signal for warm users where interactions are abundant. Concretely, we instantiate this backbone as a 2-layer LightGCN with 64-dimensional embeddings and compute a dot-product score

$$s_{cf}(u, i) = \langle \mathbf{e}_u, \mathbf{e}_i \rangle.$$

The collaborative backbone is LightGCN [37] and is trained with the Bayesian Personalized Ranking (BPR) objective [38] on sampled triplets (u, i^+, i^-) . We optimize with Adam [39] (learning rate 10^{-3} , 50 epochs, batch size 8192) and sample one negative per positive.

4.2. Semantic atoms

To ground recommendation and explanation in explicit knowledge, we represent each item by a set of semantic

atoms. A semantic atom is an interpretable unit of meaning extracted from item metadata or descriptions. For datasets with rich text (e.g., Video_Games product descriptions), we extract atoms using a large language model (DeepSeek Chat, deepseek-chat) via the DeepSeek chat-completions API. The full prompt templates are shown in Figure 2. For each item, we choose a single text description from available sources (preferring TMDB overview when present and sufficiently long, otherwise a Wikipedia summary) and prompt the model to return 12–18 plot-grounded noun phrases (1–5 words) in lowercase, formatted as a single-line JSON object with schema $\{\text{"atoms": ["string"]}\}$ (temperature 0.3, max tokens 512). The prompt explicitly forbids generic genre-level terms and discourages person names; we further postprocess model outputs by normalizing whitespace and punctuation, deduplicating atoms, removing banned generic phrases, filtering single-token proper names, and enforcing a maximum of 8 tokens per atom. We require at least 8 atoms after postprocessing and retry up to three times with exponential backoff on transient failures. We store extracted atoms as a JSONL artifact per item (including the source text, extracted atoms, provider/timestamp metadata, and an ok/error flag) to support reproducibility and auditing. All downstream recommendation experiments use only these frozen artifacts and never call the API online. In the revision artifacts used for this paper, atom extraction success rates are: Video_Games >99%, Anime 99.6%, and Yelp2018 100%.

To assess the quality of extracted atoms, we conduct an independent quality audit on a stratified random sample of 100 items per dataset (random seed 42; ~1680 atoms for Video_Games, 1739 for Anime, 1426 for Yelp2018). For each atom, we judge whether it is supported by the item's source material: for Video_Games, support requires the atom to correspond to a gameplay mechanic, genre feature, or platform attribute documented in the item description; for Anime, support requires correspondence with plot or synopsis content; for Yelp2018, support requires the atom to correspond to a documented attribute in the business knowledge graph record. Precision is computed as the fraction of supported atoms per item. Results are summarized in the audit report below, which confirms that the majority of extracted atoms are grounded in item-level evidence and suitable as an auditable semantic interface.

To construct a consistent atom vocabulary, we normalize atom strings, deduplicate atoms within each item, and build a frequency-truncated vocabulary with minimum frequency 2 and maximum size 5000. Each item i is then represented as a set of atom indices $\mathcal{A}(i)$, which we materialize as a padded tensor and a binary mask indicating whether an item has any atoms.

We treat semantic atoms as an explicit semantic interface extracted from text or structured attributes, rather than as a substitute for a curated knowledge graph. Their purpose in this work is to expose auditable semantic factors that can participate in scoring and perturbation-based diagnostics,

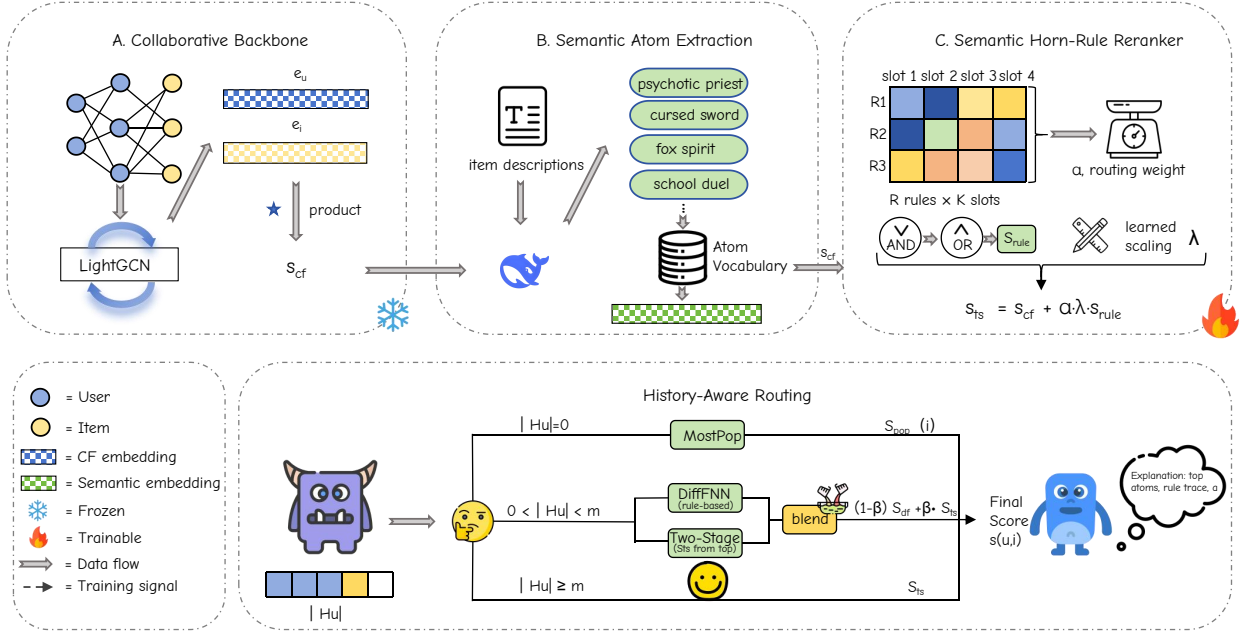


Figure 1: Overall framework of the proposed cold-start aware neuro-symbolic recommender. Top: the two-stage architecture comprising shared components—the collaborative backbone (A), semantic atom extraction (B), and semantic Horn-rule reranker (C)—that are reused across all user regimes. Bottom: history-aware routing that selects the scoring pathway based on user interaction history $|\mathcal{H}_u|$. Explanations are derived from the same atoms and rules that participate in scoring, enabling mechanism-linked auditing.

enabling mechanism-linked analysis of the semantic pathway.

4.3. Semantic horn-rule reranker

The second stage reranks candidate items using a differentiable horn-rule model that operates over semantic atoms. We first construct a user-level semantic representation by aggregating the atoms of items the user interacted with in training. Let $\mathbf{v}_a$ denote the embedding of atom a and let $c_{u,a}$ be the (sparse) count of atom a in user u 's history. We compute

$$\mathbf{z}_u = \frac{\sum_a c_{u,a} \mathbf{v}_a}{\max(\sum_a c_{u,a}, 1)}.$$

Given a candidate item i with atom embeddings $\{\mathbf{v}_a : a \in \mathcal{A}(i)\}$, we instantiate R fuzzy Horn rules, each with K slots (we use $R = 32$ and $K = 4$). For each rule r and slot k , we learn a slot query vector and compute an attention-weighted slot value that measures how well the item atoms match the user semantics. We then compute a fuzzy conjunction across slots and aggregate rules with a noisy-OR to produce a semantic rule score $s_{\text{rule}}(u, i) \in [0, 1]$. Because this computation explicitly factors through atom-level matches and a small set of learned rules, it supports structured explanations by reporting the most influential atoms and rules.

This reranker provides a semantic scoring pathway that makes atom-to-rule-to-score tracing possible. Its empirical contribution to final ranking varies by dataset: the semantic term can contribute materially to ranking when the routing

weight is high and user histories are sparse, but its primary function across all settings is to expose an auditable semantic interface rather than to universally boost accuracy.

4.4. History-aware routing

User heterogeneity is handled through a routing mechanism that selects or blends scoring pathways based on user history length. Let m be a cold-start threshold. We use two main scoring functions: a rule-based score from DiffFNN [15], denoted $s_{\text{df}}(u, i)$, and a Two-Stage score $s_{\text{ts}}(u, i)$ that combines collaborative filtering with semantic horn-rule reasoning.

The Two-Stage score adds a gated semantic rule term to a collaborative base:

$$s_{\text{ts}}(u, i) = s_{\text{cf}}(u, i) + \alpha(u, i) \lambda s_{\text{rule}}(u, i),$$

where $\lambda = \lambda_{\min} + \text{softplus}(\lambda_{\text{raw}})$ is a learned scaling factor, and $\alpha(u, i)$ is a history-aware gate. We define the user's semantic evidence as $d_u = \sum_a c_{u,a}$ (the total atom count in the user's training history), and compute:

$$\alpha(u, i) = \mathbb{I}[i \text{ has atoms}] \cdot \sigma(w \log(1 + d_u) + b) \cdot \mathbb{I}[d_u > 0],$$

where w and b are learned parameters. Thus, α controls how strongly semantic rules influence the final score: it is zero when the item has no atoms or the user has no semantic evidence, and increases with user evidence otherwise.

We then apply history-aware routing across scoring pathways. Our main method (**Ours**) uses a two-stage routing:

$$s(u, i) = \begin{cases} (1 - \beta) s_{\text{df}} + \beta s_{\text{ts}} & |\mathcal{H}_u| \leq m \\ s_{\text{ts}}(u, i) & |\mathcal{H}_u| > m \end{cases} \quad (1)$$

Semantic Atom Quality Audit Report**Dataset: Video_Games**

- ✓ Mean precision: 0.72 ± 0.19
- ✓ Items $\geq 80\%$: **38/100 (38%)**

Sample supported atoms:

“open-world exploration”, “multiplayer mode”, “action-adventure gameplay”

Sample unsupported atoms:

“competitive esports”, “mobile platform”

Dataset: Anime

- ✓ Mean precision: 0.75 ± 0.25
- ✓ Items $\geq 80\%$: **57/100 (57%)**

Sample supported atoms:

“terrorist organization”, “scylla symbol”, “computer store employee”

Sample unsupported atoms:

“cybernetics”, “pc assembly instruction”

Dataset: Yelp2018

- ✓ Mean precision: 0.85 ± 0.17
- ✓ Items $\geq 80\%$: **67/100 (67%)**

Sample supported atoms:

“restaurant or bar establishment”, “casual dress code”, “casual dining establishment”

Sample unsupported atoms:

“drive-thru service”, “specific regional cuisine”

Verdict:

- ✓ Majority of atoms are grounded in item-level evidence
- ✓ Unsupported atoms are primarily generic inferences (e.g., “accessible by car”), not factual errors

Protocol: Item-level relevance verification · 100 items/dataset (seed=42)

where $\beta = 0.5$ is a fixed symmetric blending coefficient. We intentionally use a simple symmetric blend to keep the routing policy transparent and avoid introducing an additional learned router whose behavior would itself require separate analysis. This design choice prioritizes interpretability over potential per-dataset tuning gains, and β can be adjusted as a hyperparameter when a held-out development set is available. A natural concern is whether s_{df} and s_{ts} operate on comparable score scales. Both components are trained with the same BPR pairwise ranking objective over the same item set and candidate pool, so their outputs function as ordinal ranking signals over a shared set of candidates rather than as calibrated probability estimates. This score-space interpolation over two ordinal ranking signals evaluated on the same candidate pool and does not require exact score-scale matching; we verify that the mixing does not degrade ranking quality relative to each component alone in the ablation study (Table 3). Cold users receive this blend of DiffFNN (which relies on predefined rule-based atoms) and Two-Stage scores, while warm users use the full Two-Stage model.

We sweep $\beta \in \{0.0, 0.1, 0.3, 0.5, 0.7, 0.9, 1.0\}$ on overall NDCG@10 (1 seed). On Video_Games and Anime, performance shows an inverted-U pattern peaking at $\beta=0.5$ (VG: 0.434, Anime: 0.813), with smooth degradation at extreme values ($\beta=0$: 0.402/0.733; $\beta=1$: 0.421/0.809). On Yelp2018, performance increases monotonically with β ($\beta=0$: 0.548; $\beta=0.5$: 0.621; $\beta=1$: 0.624), reflecting the relative weakness of DiffFNN on this dataset; however, the gap between $\beta=0.5$

and $\beta=1$ is negligible ($<0.5\%$ relative NDCG), so a fixed $\beta=0.5$ incurs minimal cost. Setting $\beta=0$ reduces the cold pathway to DiffFNN alone, while $\beta=1$ uses only the Two-Stage scorer. On VG and Anime, the consistent drop at both extremes confirms that the two components capture complementary cold-start signals. We note that the performance surface is relatively flat near $\beta=0.5$ ($\beta \in [0.3, 0.7]$ yields $<2\%$ relative NDCG variation on all datasets), so the routing policy is not sensitive to the precise choice of β . Our full system (**Ours3Stage**) extends this policy with a popularity fallback for super-cold users:

$$s(u, i) = \begin{cases} s_{\text{pop}}(i) & |\mathcal{H}_u| \leq m_s \\ (1 - \beta) s_{df} + \beta s_{ts} & m_s < |\mathcal{H}_u| \leq m \\ s_{ts}(u, i) & |\mathcal{H}_u| > m \end{cases} \quad (2)$$

where $s_{\text{pop}}(i)$ is the item popularity (number of training interactions) and m_s is the super-cold threshold (we set $m_s = 2$ for Video_Games and $m_s = 0$ for Anime and Yelp2018). This variant is particularly useful on datasets like Video_Games where a substantial fraction of test users have very sparse training interactions (≤ 2 interactions for 18.7% of users).

4.5. Design rationale

Several design choices merit explicit justification. First, we use a two-stage architecture rather than joint end-to-end training because our priority is a modular design that isolates the collaborative backbone from the semantic audit

System: You extract semantic atoms from an item description / plot summary for a recommender dataset. Atoms MUST be **specific, plot-grounded noun phrases (1–5 words)**. Focus on unique story elements, concrete roles, key objects, and distinctive settings. Avoid generic terms like: *romance, comedy, drama, protagonist, story, journey, conflict, relationship*. Avoid vague phrases like: *unexpected events, personal growth, emotional journey, difficult choices*. Prefer role-based phrases over character names. DO NOT output person names. Do not output abstract items like ‘better life’, ‘obstacles’, or ‘self-discovery’. Use ONLY the provided plot; do not invent facts. Return strictly valid JSON in ONE LINE. All atoms MUST be lowercase.

User Input:

```
{
  "title": "...", "year": "...", "genres": "...",
  "synopsis_source": "...", "synopsis": "...",
  "baseline_atoms": [...],
  "task": "Extract 12–18 unique semantic atoms.",
  "output_json_schema": {"atoms": ["string"]}
}
```

Output: {"atoms": ["school rooftop duel", "cursed sword", "fox spirit guardian", ...]}

Video_Games / Anime Semantic Atoms Prompt

System: You are an expert at **analyzing business characteristics** and extracting semantic atoms.

User Input:

```
Business Information (from Knowledge Graph):
- Item ID: ... - Num attributes: ... - Num relations: ...
Key Attributes: [city: Phoenix, category: Restaurants, ...]
```

Requirements:

1. Each atom should be a short phrase (2–5 words).
2. Infer business type, atmosphere, services, and customer experience from the attributes.
3. Be specific and descriptive (e.g., “casual dining”, “outdoor seating available”).
4. Focus on customer-relevant features.
5. Avoid just repeating attribute names — interpret what they mean for customers.

Task: Based on these attributes, **extract 8–15 semantic atoms** that describe what kind of business this is.

Output: ["casual dining", "family-friendly", "outdoor seating", "accepts reservations", ...]

Business Semantic Atoms Prompt (Yelp2018)

Figure 2: Semantic atom extraction prompt templates. The *Anime / Video_Games* prompt (top) extracts 12–18 semantically grounded atoms from item descriptions, while the *Yelp* prompt (bottom) infers 8–15 customer-relevant features from knowledge graph attributes. The Anime and Video_Games prompts share the same structure, differing only in the input field names (synopsis vs. description).

layer, facilitating independent analysis of each component’s contribution. Second, we use inference-time routing based on history length rather than a learned router because the goal is a transparent, lightweight, and interpretable regime policy; a learned router would introduce additional parameters whose behavior would require separate faithfulness analysis. Third, semantic atoms remain valuable even when they do not universally boost accuracy because they provide the foundation for mechanism-linked diagnostics: the perturbation protocol and case studies depend on having explicit, inspectable semantic factors that participate in scoring. Fourth, the super-cold branch uses popularity as a stable fallback when collaborative evidence is extremely sparse ($|\mathcal{H}_u| \leq 2$); this is especially important on Video_Games, where 18.7% of evaluation users satisfy this condition, and provides measurable benefits on Yelp2018 (2.8%); on Anime the super-cold fraction (1.2%) is too small to affect aggregate metrics.

4.6. Training and optimization

We train the collaborative backbone and the semantic reranker in two stages. The LightGCN backbone is optimized with BPR loss using one uniformly sampled negative item per observed positive interaction and the Adam optimizer (learning rate 10^{-3} , 50 epochs, batch size 8192). After training LightGCN, we freeze its embeddings and train the SemanticHornReranker on the same BPR triplets, using the LightGCN dot-product as s_{cf} and learning all reranker parameters (atom embeddings, rule/slot parameters, λ_{raw} , and gate parameters w, b) with Adam (learning rate 10^{-3} , 30 epochs, batch size 8192). In the reranker, we set $\lambda_{min} = 0.1$ and temperature to 1.0.

The DiffFNN component is trained separately following the configuration reported by Bartl et al. [15]: it learns fuzzy Horn rules over predefined atoms with MSE loss and L1 regularization (100 epochs, learning rate 0.01, L1 weight 0.001). Critically, once both components are trained, no regime-specific model copies are maintained; the routing policy selects among the shared trained components at inference time based on each user’s history length.

Table 1: Main results (5 seeds). Bold: best; underline: second-best; *: $p < 0.05$ vs. DiffFNN. Blue rows denote recently added strong baselines (RLMRec [43], LightGCL [44], ELCRec [49], AlphaRec [50]).

Dataset	Method	Recall			HR			NDCG		
		@5	@10	@20	@5	@10	@20	@5	@10	@20
Video_Games	DiffFNN	0.502 _(0.010)	0.616 _(0.008)	0.727 _(0.012)	0.502 _(0.010)	0.616 _(0.008)	0.727 _(0.012)	0.368 _(0.006)	0.402 _(0.007)	0.432 _(0.006)
	LightGCL	0.502 _(0.004)	0.608 _(0.004)	0.715 _(0.003)	0.502 _(0.003)	0.608 _(0.004)	0.715 _(0.003)	0.365 _(0.003)	0.395 _(0.004)	0.424 _(0.003)
	NeuMF	0.423 _(0.006)	0.529 _(0.003)	0.641 _(0.002)	0.423 _(0.006)	0.529 _(0.003)	0.641 _(0.002)	0.312 _(0.005)	0.349 _(0.003)	0.379 _(0.003)
	MostPop	0.705 _(0.001)	0.805 _(0.001)	0.894 _(0.001)	0.705 _(0.001)	0.805 _(0.001)	0.894 _(0.001)	0.551 _(0.000)	0.557 _(0.000)	0.569 _(0.000)
	Two-Stage	0.538 _(0.004)	0.649 _(0.003)	0.754 _(0.008)	0.538 _(0.004)	0.649 _(0.003)	0.754 _(0.008)	0.391 _(0.005)	0.421 _(0.004)	0.449 _(0.004)
	Ours	0.554 _(0.005)	0.669 _(0.005)	0.778 _(0.009)	0.554 _(0.005)	0.669 _(0.005)	0.778 _(0.009)	0.403 _(0.007)	0.434 _(0.004)	0.463 _(0.003)
	Ours3Stage	0.680 _(0.015)	0.776 _(0.014)	0.861 _(0.027)	0.680 _(0.015)	0.776 _(0.014)	0.861 _(0.027)	0.531 _(0.007)	0.537 _(0.003)	0.549 _(0.007)
	RLMRec	0.561 _(0.002)	0.689 _(0.003)	0.815 _(0.003)	0.561 _(0.002)	0.689 _(0.003)	0.815 _(0.003)	0.419 _(0.005)	0.460 _(0.004)	0.479 _(0.002)
	LightGCL	0.430 _(0.001)	0.538 _(0.003)	0.653 _(0.005)	0.430 _(0.001)	0.538 _(0.003)	0.653 _(0.005)	0.317 _(0.003)	0.355 _(0.002)	0.386 _(0.003)
	ELCRec	0.614 _(0.002)	0.727 _(0.002)	0.828 _(0.001)	0.614 _(0.002)	0.727 _(0.002)	0.828 _(0.001)	0.482 _(0.001)	0.519 _(0.001)	0.544 _(0.001)
	AlphaRec	0.540 _(0.002)	0.662 _(0.002)	0.779 _(0.002)	0.540 _(0.002)	0.662 _(0.002)	0.779 _(0.002)	0.411 _(0.001)	0.451 _(0.001)	0.480 _(0.001)
	Improv. vs DiffFNN	35.5%	26.0%	18.4%	35.5%	26.0%	18.4%	44.3%	33.6%	27.1%
Anime	DiffFNN	0.185 _(0.011)	0.339 _(0.025)	0.560 _(0.041)	0.905 _(0.012)	0.967 _(0.007)	0.988 _(0.004)	0.477 _(0.024)	0.493 _(0.027)	0.549 _(0.034)
	LightGCL	0.352 _(0.003)	0.556 _(0.004)	0.768 _(0.003)	0.988 _(0.001)	0.997 _(0.001)	0.999 _(0.000)	0.828 _(0.005)	0.812 _(0.004)	0.835 _(0.004)
	NeuMF	0.373 _(0.002)	0.589 _(0.002)	0.799 _(0.002)	0.992 _(0.003)	0.997 _(0.000)	0.999 _(0.001)	0.862 _(0.003)	0.849 _(0.003)	0.871 _(0.003)
	EASE	0.418 _(0.002)	0.641 _(0.002)	0.826 _(0.002)	0.997 _(0.001)	0.999 _(0.001)	0.999 _(0.000)	0.937 _(0.002)	0.923 _(0.001)	0.928 _(0.002)
	ItemKNN	0.055 _(0.001)	0.122 _(0.001)	0.249 _(0.002)	0.399 _(0.004)	0.594 _(0.008)	0.791 _(0.005)	0.145 _(0.002)	0.167 _(0.001)	0.211 _(0.002)
	MostPop	0.330 _(0.001)	0.520 _(0.003)	0.729 _(0.002)	0.975 _(0.001)	0.989 _(0.002)	0.996 _(0.001)	0.785 _(0.004)	0.765 _(0.004)	0.790 _(0.002)
	Two-Stage	0.353 _(0.002)	0.557 _(0.003)	0.769 _(0.003)	0.988 _(0.001)	0.997 _(0.001)	0.999 _(0.000)	0.830 _(0.003)	0.813 _(0.003)	0.837 _(0.002)
	Ours	0.353 _(0.001)	0.557 _(0.002)	0.769 _(0.002)	0.987 _(0.000)	0.996 _(0.000)	0.999 _(0.000)	0.829 _(0.002)	0.813 _(0.002)	0.836 _(0.002)
	Ours3Stage	0.353 _(0.001)	0.557 _(0.002)	0.769 _(0.002)	0.987 _(0.000)	0.996 _(0.000)	0.999 _(0.000)	0.829 _(0.002)	0.813 _(0.002)	0.836 _(0.002)
	RLMRec	0.361 _(0.001)	0.556 _(0.002)	0.720 _(0.002)	0.972 _(0.001)	0.973 _(0.001)	0.974 _(0.001)	0.858 _(0.001)	0.832 _(0.002)	0.824 _(0.001)
	LightGCL	0.359 _(0.004)	0.551 _(0.007)	0.712 _(0.010)	0.967 _(0.001)	0.973 _(0.001)	0.975 _(0.001)	0.848 _(0.003)	0.823 _(0.004)	0.816 _(0.007)
	ELCRec	0.623 _(0.003)	0.773 _(0.003)	0.886 _(0.004)	0.623 _(0.003)	0.773 _(0.003)	0.886 _(0.004)	0.462 _(0.003)	0.511 _(0.004)	0.540 _(0.003)
	AlphaRec	0.654 _(0.005)	0.789 _(0.006)	0.890 _(0.002)	0.654 _(0.005)	0.789 _(0.006)	0.890 _(0.002)	0.494 _(0.004)	0.538 _(0.004)	0.564 _(0.004)
	Improv. vs DiffFNN	90.8%	64.3%	37.3%	9.1%	3.0%	1.1%	73.8%	64.9%	52.3%
Yelp2018	DiffFNN	0.121 _(0.008)	0.220 _(0.013)	0.390 _(0.012)	0.429 _(0.019)	0.624 _(0.021)	0.822 _(0.009)	0.143 _(0.008)	0.180 _(0.010)	0.243 _(0.010)
	LightGCL	0.497 _(0.005)	0.692 _(0.004)	0.860 _(0.003)	0.911 _(0.003)	0.967 _(0.001)	0.990 _(0.001)	0.567 _(0.006)	0.630 _(0.005)	0.688 _(0.005)
	NeuMF	0.426 _(0.004)	0.612 _(0.005)	0.796 _(0.007)	0.867 _(0.003)	0.947 _(0.002)	0.985 _(0.001)	0.493 _(0.004)	0.553 _(0.004)	0.618 _(0.004)
	MostPop	0.258 _(0.001)	0.390 _(0.001)	0.550 _(0.001)	0.655 _(0.001)	0.799 _(0.001)	0.907 _(0.001)	0.301 _(0.000)	0.346 _(0.000)	0.403 _(0.000)
	Two-Stage	0.456 _(0.007)	0.654 _(0.006)	0.840 _(0.003)	0.885 _(0.005)	0.956 _(0.002)	0.989 _(0.001)	0.525 _(0.007)	0.627 _(0.006)	0.686 _(0.005)
	Ours	0.454 _(0.006)	0.653 _(0.006)	0.838 _(0.003)	0.884 _(0.005)	0.956 _(0.001)	0.988 _(0.000)	0.523 _(0.007)	0.624 _(0.006)	0.683 _(0.005)
	Ours3Stage	0.456 _(0.006)	0.656 _(0.006)	0.841 _(0.003)	0.886 _(0.005)	0.957 _(0.001)	0.989 _(0.000)	0.526 _(0.007)	0.631 _(0.006)	0.689 _(0.005)
	RLMRec	0.500 _(0.001)	0.658 _(0.004)	0.778 _(0.003)	0.887 _(0.002)	0.924 _(0.003)	0.942 _(0.001)	0.588 _(0.002)	0.631 _(0.002)	0.668 _(0.003)
	LightGCL	0.503 _(0.001)	0.661 _(0.001)	0.781 _(0.002)	0.890 _(0.001)	0.927 _(0.001)	0.946 _(0.001)	0.588 _(0.001)	0.632 _(0.002)	0.671 _(0.001)
	ELCRec	0.744 _(0.003)	0.868 _(0.001)	0.952 _(0.000)	0.744 _(0.003)	0.868 _(0.001)	0.952 _(0.000)	0.571 _(0.003)	0.612 _(0.002)	0.633 _(0.002)
	AlphaRec	0.643 _(0.002)	0.781 _(0.003)	0.888 _(0.002)	0.643 _(0.002)	0.781 _(0.003)	0.888 _(0.002)	0.476 _(0.003)	0.523 _(0.003)	0.550 _(0.003)
	Improv. vs DiffFNN	276.9%	198.2%	115.6%	106.5%	53.4%	20.3%	267.8%	250.6%	183.5%

4.7. Deployment considerations

Our approach is designed to expose auditability signals with limited additional inference cost. In deployment, the collaborative backbone can be used to retrieve a small candidate set, which is then reranked by the semantic horn-rule component; explanations are generated from the same intermediate quantities already computed for scoring (atoms, rule activations, and the routing weight). In our evaluation protocol, each evaluation instance is scored against a fixed candidate set shared across methods, providing a realistic and computationally tractable assessment of ranking quality.

4.8. Mechanism sensitivity evaluation

To probe mechanism sensitivity, we treat semantic atoms as mechanism-level features used by the reranker rather than as post-hoc explanation tokens. We perform counterfactual perturbations by masking or removing subsets of atoms and measure changes in scores and ranking metrics. In particular, we compare removing atoms highlighted by the explanation versus removing randomly selected atoms under the same budget. We view these perturbation results as a reproducible diagnostic of whether the reported atoms align with model sensitivity, and we complement them with qualitative case studies tracing recommendations to atoms and rules.

5. Experimental Setup

We evaluate on three public benchmarks: Video_Games 2023 [51], Anime, and Yelp2018. For each dataset, we adopt a per-user chronological holdout split. We report HR, Recall, and NDCG at $K \in \{5, 10, 20\}$, and we present results for overall users as well as cold and warm user subsets derived from the history quantile threshold [40]. Video_Games (94,762 users, 25,612 items, 814,586 interactions) is a recent, large-scale dataset from the Amazon Reviews 2023 collection.

To ensure reliable conclusions, we repeat experiments with five random seeds and report mean and standard deviation. We compare against a diverse set of baselines including popularity, collaborative models, and a neuro-symbolic baseline. In the revised version, we additionally include four recent strong baselines: RLMRec [43] (LLM-enhanced collaborative filtering, WWW 2024), LightGCL [44] (SVD-augmented graph contrastive learning, ICLR 2023), ELCRec [49] (SASRec-based contrastive cold-start, NeurIPS 2024), and AlphaRec [50] (contrastive LightGCN with LLM embeddings, ICLR 2025 Oral). These baselines serve two different purposes in our study. Strong black-box recommenders such as NeuMF, EASE, RLMRec, LightGCL, ELCRec, AlphaRec, and LightGCN calibrate overall competitiveness, while DiffFNN and our internal ablations serve as mechanism-level references for evaluating whether the proposed architectural decomposition improves over an existing neuro-symbolic design. For reporting clarity, all main results are shown as mean $\pm$ standard deviation over five seeds; unless otherwise noted, percentage improvements reported in the main text and tables are relative to DiffFNN. Our

main focus is to quantify the benefit of combining history-aware routing with semantic horn-rule reasoning and to demonstrate that this combination provides a mechanism-linked diagnostic interface for auditing and debugging recommendations.

6. Results and Discussion

Table 1 shows that Ours3Stage significantly improves over DiffFNN on Recall and NDCG across all three datasets and five random seeds. On Video_Games, Ours3Stage achieves $\text{NDCG@10} = 0.537$, outperforming all recent learning-based baselines including ELCRec (0.519, NeurIPS 2024), RLMRec (0.460, WWW 2024), AlphaRec (0.451, ICLR 2025), and LightGCL (0.355, ICLR 2023); only the popularity baseline MostPop (0.557) is higher, which is expected given Video_Games's long-tail distribution. On Yelp2018, Ours3Stage (0.631) is comparable to the strongest baselines RLMRec (0.631) and LightGCL (0.632) in mean NDCG@10 , with overlapping confidence intervals across seeds. On Anime, strong collaborative methods (EASE, NeuMF) dominate, but our method still outperforms both ELCRec (0.511) and AlphaRec (0.538). These results demonstrate that our approach offers competitive accuracy on recent, large-scale benchmarks while simultaneously providing an auditable semantic pathway that none of the black-box baselines offer.

Figure 3 provides a regime-level view of the same story.

Relative to DiffFNN, Ours3Stage moves upward and to the right on the cold/warm plane, showing that routing improves robustness across user regimes rather than only shifting a single aggregate mean. This interpretation is especially clear on Video_Games and Yelp2018, where the routed policy yields stronger cold-side gains.

Beyond mean performance, we also inspect user-level behavior to support the system-level story behind history-aware routing. On Video_Games, this improvement aligns with the substantial super-cold population (18.7% of evaluation users have $|\mathcal{H}_u| \leq 2$), where the popularity fallback stabilizes rankings for users with minimal collaborative evidence.

The gains also differ systematically across datasets. On Video_Games, the largest improvement comes from frequent activation of the super-cold branch (18.7% of users), which explains why Ours3Stage substantially exceeds Ours (+0.103 NDCG@10). On Anime and Yelp2018, the super-cold branch activates less frequently (1.2% and 2.8% of users respectively); the gain is negligible on Anime (where the super-cold population is too small to affect aggregate metrics) but measurable on Yelp2018 (NDCG@10 : +0.007), indicating that the popularity fallback provides consistent benefits when the super-cold fraction is non-trivial.

The semantic pathway contributes differentially across datasets: on Yelp2018 it contributes more strongly to the final score, which helps explain why ranking-level semantic sensitivity becomes visible there, while on Anime and Video_Games the semantic contribution is more modest but remains inspectable. Beyond accuracy, the semantic

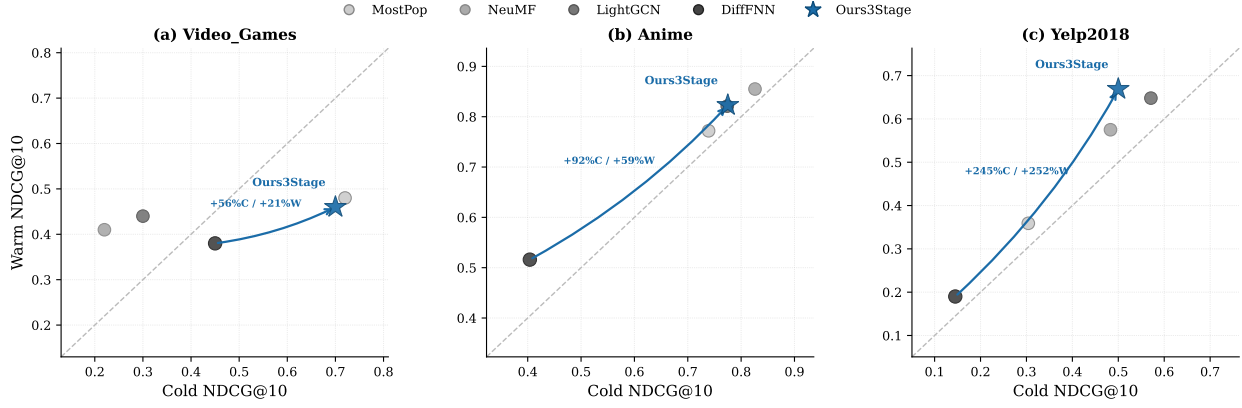


Figure 3: Cold vs Warm performance scatter (NDCG@10). Dashed line shows $y=x$. Arrows connect DiffFNN to Ours3Stage, and Δ labels report percentage improvement on the cold and warm splits.

Table 2

Cold/warm split performance (NDCG@10, 5 seeds). We report a representative subset of methods.

	Video_Games		Anime		Yelp2018	
Method	Cold	Warm	Cold	Warm	Cold	Warm
MostPop	0.720 _(0.002)	0.480 _(0.000)	0.739 _(0.011)	0.772 _(0.004)	0.304 _(0.001)	0.359 _(0.000)
LightGCN	0.300 _(0.005)	0.440 _(0.006)	0.774 _(0.010)	0.821 _(0.004)	0.571 _(0.005)	0.648 _(0.005)
NeuMF	0.220 _(0.005)	0.410 _(0.006)	0.826 _(0.006)	0.855 _(0.003)	0.483 _(0.005)	0.575 _(0.004)
EASE	—	—	0.907 _(0.002)	0.928 _(0.002)	—	—
RLMRec	0.273 _(0.005)	0.549 _(0.011)	0.720 _(0.003)	0.861 _(0.002)	0.473 _(0.003)	0.678 _(0.003)
LightGCL	0.260 _(0.005)	0.400 _(0.002)	0.738 _(0.011)	0.845 _(0.003)	0.506 _(0.003)	0.670 _(0.001)
DiffFNN	0.450 _(0.013)	0.380 _(0.006)	0.404 _(0.026)	0.516 _(0.027)	0.145 _(0.009)	0.190 _(0.010)
Two-Stage	0.360 _(0.007)	0.450 _(0.005)	0.775 _(0.008)	0.823 _(0.004)	0.507 _(0.008)	0.662 _(0.006)
Ours	0.400 _(0.004)	0.450 _(0.005)	0.775 _(0.008)	0.823 _(0.004)	0.500 _(0.012)	0.660 _(0.006)
Ours3Stage	0.700 _(0.015)	0.460 _(0.006)	0.775 _(0.008)	0.823 _(0.004)	0.500 _(0.012)	0.669 _(0.006)

atoms therefore provide a concrete handle for analysis and verification. Sensitivity diagnostics based on atom perturbations provide a reproducible check that links reported atoms to measurable changes in the reranker’s scoring behavior. Qualitative case studies further illustrate how the model can surface user-relevant semantic atoms and learned rule patterns that jointly justify the recommendation, making the reasoning process inspectable and actionable.

Three boundary conditions should be stated before moving to the finer analyses. First, the popularity baseline MostPop still outperforms our method on Video_Games overall accuracy due to the dataset’s extreme long-tail distribution, though Ours3Stage surpasses all learning-based baselines. Second, all methods are evaluated on the same held-out instances and candidate sets, providing a fair comparison. Third, ranking-level semantic sensitivity is weaker but detectable on Video_Games and Anime under the current setup, with effect sizes growing with removal budget, while Yelp2018 shows the strongest signal consistent with its higher routing weight. Figures 4 and 5 further visualize these trends across methods and K values.

Table 2 reveals an important nuance that overall accuracy alone does not capture. On Video_Games cold users,

Ours3Stage (0.700) dramatically outperforms all learning-based baselines: LightGCN (0.300), NeuMF (0.220), RLMRec (0.273), LightGCL (0.260), DiffFNN (0.450), and Two-Stage (0.360), and is competitive with MostPop (0.720). On Anime cold users, EASE (0.907) is the strongest, but Ours3Stage (0.775) outperforms LightGCL (0.738) and RLMRec (0.720), and is competitive with LightGCN (0.774). On Yelp2018, Ours3Stage (0.500) outperforms MostPop (0.304), RLMRec (0.473), and NeuMF (0.483), and is competitive with LightGCL (0.506) and LightGCN (0.571). Notably, on Video_Games warm users, RLMRec (0.549) is the strongest due to its LLM-enhanced embeddings, but our method still surpasses LightGCN (0.440), NeuMF (0.410), and LightGCL (0.400). These results show that the accuracy gap between our method and the strongest warm-side baselines is a warm-user phenomenon where LLM-rich methods extract more signal from extensive histories. In contrast, when history is sparse, these same models degrade sharply because their embedding-heavy design has no meaningful fallback, whereas our routing mechanism provides a robust cold-user pathway. This supports our positioning: Ours3Stage provides (i) competitive or superior cold-user performance across all three datasets, (ii) a transparent

Table 3

Ablation study (NDCG, overall split, 5 seeds). Bold: best per column.

Group	Method	Video_Games		Anime		Yelp2018	
		@10	@20	@10	@20	@10	@20
Baselines	DiffFNN	0.402	0.432	0.493	0.549	0.180	0.243
	Two-Stage (full)	0.421	0.449	0.813	0.837	0.627	0.686
Component ablations	w/o semantic	0.395	0.424	0.812	0.835	0.630	0.688
	w/o gate	0.403	0.428	0.807	0.831	0.627	0.686
	semantic only	0.211	0.254	0.215	0.234	0.129	0.164
	shuffled atoms	0.291	0.328	0.812	0.835	0.620	0.679
Routing	Ours	0.434	0.462	0.813	0.836	0.624	0.683
	Ours3Stage	0.537	0.549	0.813	0.836	0.631	0.689

Note: w/o semantic removes s_{rule} ; w/o gate removes α ; semantic only removes s_{cf} ; shuffled atoms randomize atom-item assignment.

routing policy, and (iii) an auditable semantic pathway for every recommendation.

At the same time, the results highlight a broader point: in production recommendation, the best-performing black-box model is not always the most deployable one. When auditability, user trust, and controllability matter—as they increasingly do in regulated domains and high-stakes applications—a knowledge-based interface to recommendation decisions becomes a system-level advantage. Our framework is therefore best understood as an auditability-oriented system design rather than as a pure accuracy-maximization model, and its mechanism-linked sensitivity protocol can be adapted to other neuro-symbolic architectures.

6.1. Ablation study

We conduct extensive ablations to disentangle the contribution of routing and the semantic reranking components. Table 3 reports NDCG results on the overall split for all three datasets, averaged over five random seeds. The central message of this table is not merely that some components help more than others; rather, it reveals the system decomposition that underpins the entire paper. Routing accounts for most of the ranking gains, whereas the semantic term primarily enables an auditable semantic pathway. In particular, Two-Stage (full) and its semantic ablations are often close in NDCG, while Ours3Stage is the variant that yields the clearest gain over DiffFNN. This supports the claim that semantic reranking is not the main source of NDCG gains, even though it remains essential as the mechanism-linked semantic interface. Full results with additional metrics (HR, Recall) and K values are provided in the supplementary material.

Importantly, we separate two complementary contributions that should not be conflated. First, history-aware routing (including the super-cold popularity fallback in Ours3Stage) is the primary driver of accuracy gains: it stabilizes cold-start behavior and contributes substantially to the improved NDCG across user regimes. Second, the semantic atoms and horn-rule reranker contribute primarily to auditability rather than accuracy: they expose mechanism-linked auditing signals (atoms, rules, and routing weights)

that support debugging and verification. This asymmetric decomposition is deliberate rather than accidental. The ablations in Table 3 confirm it directly: removing routing degrades accuracy substantially, while removing the semantic term has limited impact on NDCG across settings but removes the knowledge-grounded mechanism interface that enables qualitative tracing, perturbation testing, and manual audit. At the same time, because the largest gains arise from routing, the most direct missing control would be a stronger family of targeted routing heuristics, such as using MostPop for zero-history users together with a collaborative-only model for the remaining users. Our current results therefore establish the value of the overall architecture relative to DiffFNN and the reported ablations, but they do not prove optimality against every simple regime heuristic.

To directly address this point, Table 4 compares Ours3Stage against HeuristicRoute, a simple cold-aware heuristic that routes cold users (history $< m$) to MostPop and warm users to LightGCN. This represents a strong non-semantic routing baseline because it combines the best-performing popularity method for cold users with the best collaborative model for warm users, without any semantic component. Ours3Stage consistently outperforms HeuristicRoute on all three datasets in overall NDCG (Video_Games: 0.537 vs 0.530; Anime: 0.813 vs 0.805; Yelp2018: 0.631 vs 0.568), with particularly large gains on Yelp2018 cold users (+19.6 NDCG@10 points). On Video_Games, the margin is small because MostPop is extremely effective in this long-tail dataset, giving HeuristicRoute a strong cold-side anchor (0.720); nevertheless, Ours3Stage provides a competitive cold-user pathway (0.700) while also exposing an auditable semantic pathway that HeuristicRoute entirely lacks. These results confirm that semantic routing provides both aggregate accuracy gains and structural diagnostic value over simpler heuristic alternatives.

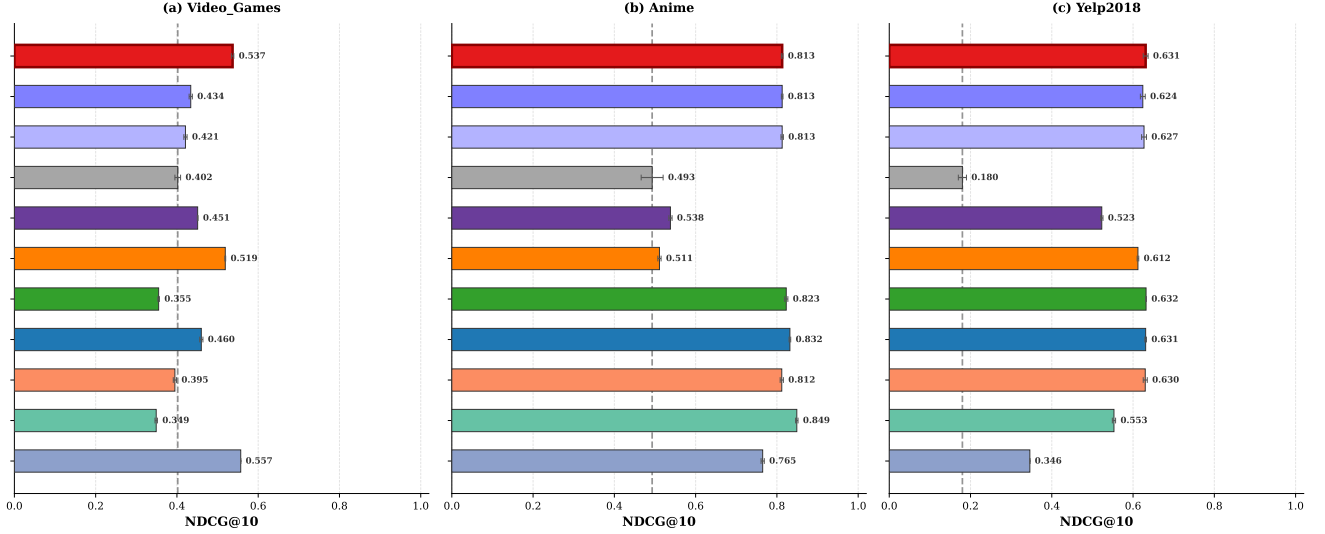
6.2. Cold-start sensitivity

Cold-start is not a single regime but a continuum: the same user can move from having no history to having a handful of interactions and eventually becoming warm. To test whether our approach behaves robustly across this

Table 4

Routing strategy comparison (NDCG@10, 5 seeds). HeuristicRoute: cold→MostPop, warm→LightGCN.

Method	Video_Games			Anime			Yelp2018		
	Overall	Cold	Warm	Overall	Cold	Warm	Overall	Cold	Warm
MostPop	0.557 _(0.000)	0.720 _(0.002)	0.480 _(0.000)	0.765 _(0.004)	0.739 _(0.011)	0.772 _(0.004)	0.346 _(0.000)	0.304 _(0.001)	0.359 _(0.000)
LightGCN	0.395 _(0.004)	0.300 _(0.005)	0.440 _(0.006)	0.812 _(0.004)	0.774 _(0.010)	0.821 _(0.004)	0.630 _(0.005)	0.571 _(0.005)	0.648 _(0.005)
HeuristicRoute	0.530 _(0.003)	0.720 _(0.002)	0.440 _(0.006)	0.805 _(0.004)	0.739 _(0.011)	0.821 _(0.004)	0.568 _(0.004)	0.304 _(0.001)	0.648 _(0.005)
Ours3Stage	0.537 _(0.003)	0.700 _(0.015)	0.460 _(0.006)	0.813 _(0.002)	0.775 _(0.008)	0.823 _(0.004)	0.631 _(0.006)	0.500 _(0.012)	0.669 _(0.006)

**Figure 4:** NDCG@10 comparison for representative methods across datasets (overall split, 5 seeds). Error bars: standard deviation. Values are plotted from Table 1; EASE and ItemKNN are omitted for visual clarity.

continuum, we perform a sensitivity analysis where we vary the history threshold m used to define cold users and re-evaluate performance on users with $|\mathcal{H}_u| \leq m$. This analysis is computed directly from the saved per-user artifacts and therefore isolates how the scoring function responds to progressively richer evidence, without changing the trained parameters.

Table 5 reports NDCG@10 under several thresholds for each dataset. The trends are dataset dependent, reflecting differences in sparsity and in the distribution of user histories. Nevertheless, across datasets we observe that the routed variants remain stable as the definition of cold-start changes, rather than overfitting to a single hand-picked threshold. On Video_Games, performance decreases as m grows because the cold pool progressively includes warmer users for whom the popularity fallback is less competitive; crucially, however, the relative advantage over DiffFNN remains consistent ($\approx +50\%$) across thresholds, confirming that routing does not overfit to the operational threshold. This is practically important because real systems rarely expose a universal notion of “cold”, and a deployable method should not be sensitive to small shifts in how cold users are defined.

6.3. Fine-grained user-bin and item-popularity proxy analysis

To complement the binary cold/warm partition, we report NDCG@10 across finer user-history bins and an item-popularity proxy analysis in Table 6, computed from one seed.

Table 6(a) partitions evaluation users by training-history length. On Video_Games, Ours3Stage achieves the strongest performance on super-cold users (bin 0–2) thanks to the popularity fallback, while matching Two-Stage for warm bins. On Yelp2018, Two-Stage and Ours3Stage dominate across all bins, with particularly large gains over MostPop and DiffFNN for low-interaction users.

Table 6(b) groups users by the median popularity of their test items and should be interpreted as an item-coldness proxy rather than as a full new-item benchmark. Under the current temporal splits, truly unseen test items are rare or absent (Video_Games: 142, 0.17%; Anime: 4, 0.01%; Yelp2018: 0). Because a dedicated new-item evaluation would therefore be statistically weak in the present split, we use item-popularity bins as a more stable approximation to item-side coldness. Users whose test items are cold (low-popularity) are harder to serve because collaborative signals are weak for rarely-interacted items. MostPop expectedly collapses on the lowest-popularity bins (e.g., 0.118 on

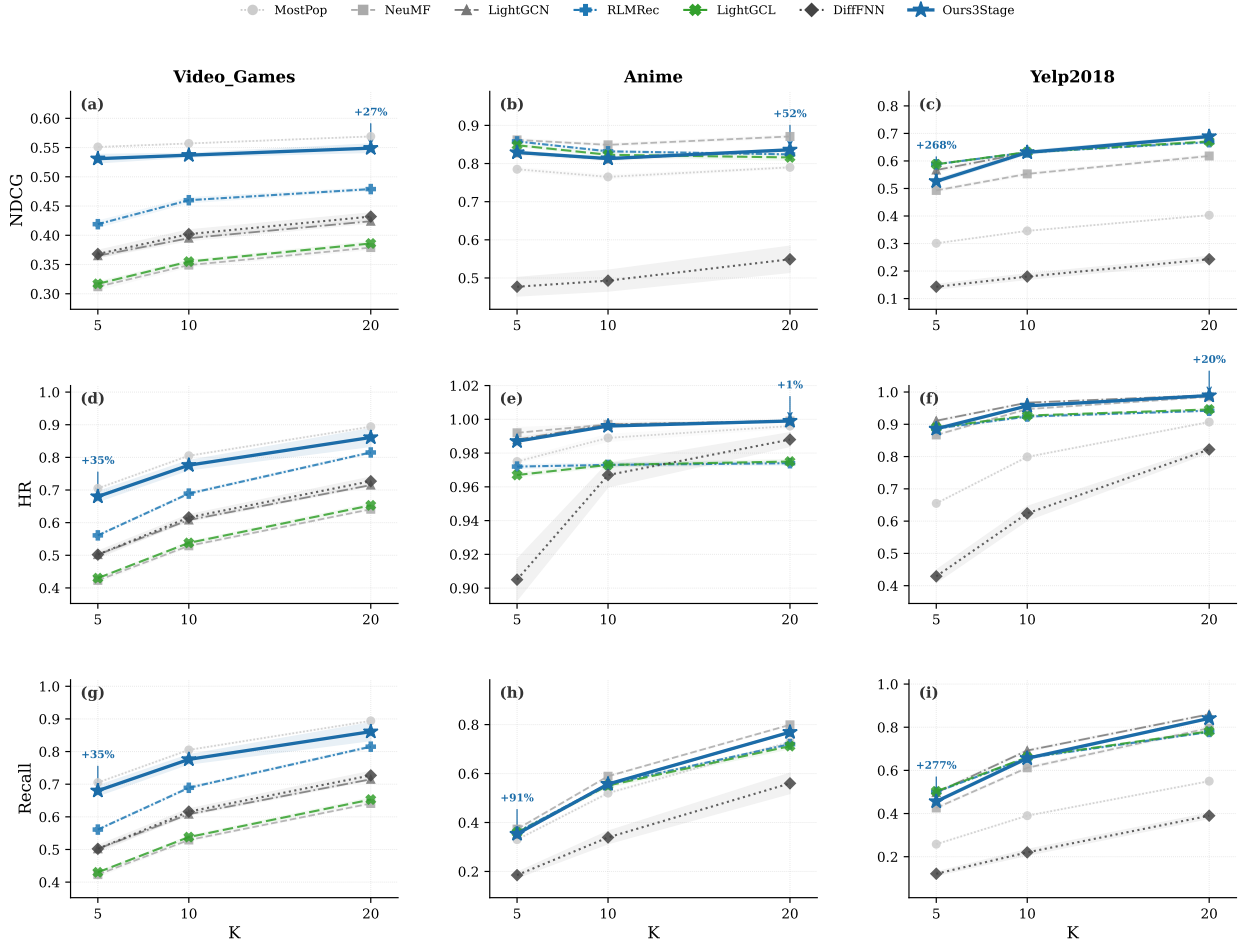


Figure 5: Performance across $K \in \{5, 10, 20\}$ (overall split, 5 seeds). Rows: metrics; columns: datasets. Shaded bands: ± 1 std. Percentage annotations, where shown, are computed relative to DiffFNN using the same values reported in Table 1.

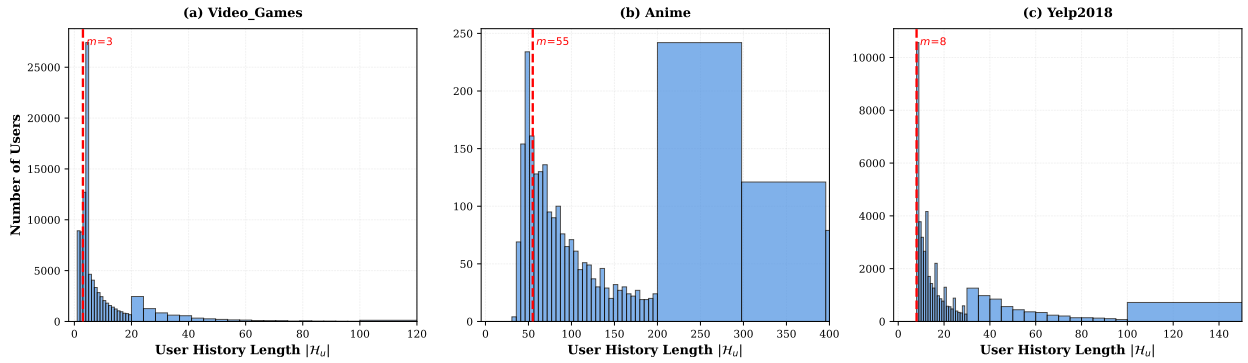


Figure 6: Distribution of evaluation-user history lengths. Red dashed lines mark cold-start threshold m ; annotations show percentage of super-cold, cold, and warm users under the quantile-based protocol.

Video_Games 5–19, 0.030 on Anime 5–19) because it can only recommend already-popular items. In contrast, Two-Stage and Ours3Stage maintain strong performance even on low-popularity items—for example, on Video_Games bin 5–19, Ours3Stage achieves 0.247 versus MostPop’s 0.118 and on Yelp2018 bin 0–4, Ours3Stage achieves 0.264 versus

MostPop’s 0.054—because the semantic pathway can leverage item-level atom features that are available regardless of interaction frequency. This supports item-side robustness under scarce interaction frequency, although it should not be read as a substitute for a dedicated new-item benchmark (see Limitation 8).

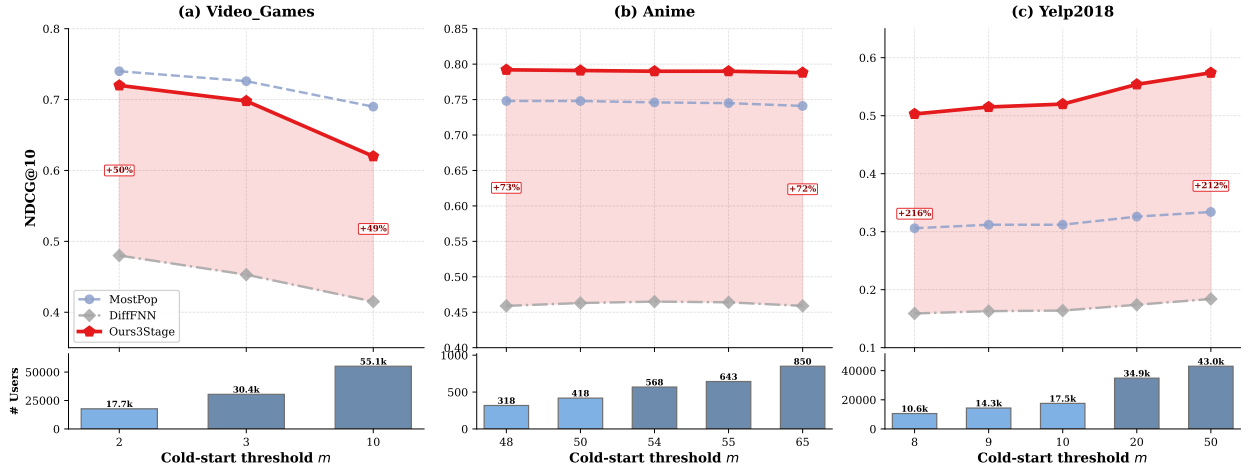


Figure 7: Cold-start sensitivity across threshold settings (NDCG@10, 1 seed). Upper panels show NDCG@10 at each threshold m ; lower bars show the number of cold users included under each threshold.

Table 5

Cold-start sensitivity analysis (NDCG@10, 1 seed, full test set). Varying history threshold m for cold user definition. n : number of cold users.

Dataset	m	n	%cold	MostPop	DiffFNN	Two-Stage	Ours	Ours3Stage
Video_Games	2	17720	18.7%	0.740	0.480	0.310	0.370	0.720
	3	30420	32.1%	0.726	0.453	0.331	0.382	0.698
	10	55113	58.2%	0.690	0.415	0.375	0.410	0.620
Anime	48	318	10.1%	0.748	0.459	0.786	0.786	0.792
	50	418	13.3%	0.748	0.463	0.785	0.785	0.791
	54	568	18.1%	0.746	0.465	0.784	0.784	0.790
	55	643	20.5%	0.745	0.464	0.784	0.784	0.790
	65	850	27.1%	0.741	0.459	0.782	0.782	0.788
Yelp2018	8	10563	24.0%	0.306	0.159	0.500	0.498	0.503
	9	14341	32.6%	0.312	0.163	0.511	0.510	0.515
	10	17529	39.8%	0.312	0.164	0.516	0.515	0.520
	20	34860	79.2%	0.326	0.174	0.550	0.549	0.554
	50	43025	97.7%	0.334	0.184	0.569	0.569	0.574

Table 6

Fine-grained performance breakdown (NDCG@10, 1 seed). Bold: best per column.

(a) By user-history bin. Bin boundaries denote training interaction count.											
Method	Video_Games				Anime			Yelp2018			
	0–2	3–5	6–10	10+	11–50	51–100	100+	1–10	11–50	51–100	100+
MostPop	0.738	0.714	0.647	0.521	0.756	0.734	0.792	0.312	0.349	0.486	0.668
DiffFNN	0.476	0.431	0.388	0.367	0.395	0.425	0.570	0.164	0.197	0.325	0.507
Two-Stage	0.309	0.361	0.417	0.473	0.780	0.767	0.843	0.516	0.606	0.789	0.908
Ours3Stage	0.738	0.682	0.469	0.475	0.794	0.767	0.843	0.524	0.606	0.789	0.908

(b) By item popularity bin. Users are grouped by median popularity of their test items.										
Method	Video_Games			Anime			Yelp2018			
	5–19	20–99	100+	5–19	20–99	100+	0–4	5–19	20–99	100+
MostPop	0.118	0.476	0.683	0.030	0.441	0.790	0.054	0.115	0.346	0.733
DiffFNN	0.093	0.317	0.453	0.119	0.360	0.498	0.052	0.110	0.214	0.285
Two-Stage	0.183	0.378	0.492	0.034	0.638	0.818	0.263	0.460	0.605	0.747
Ours3Stage	0.247	0.491	0.568	0.041	0.651	0.824	0.264	0.467	0.612	0.752

6.4. Case study: tracing recommendations to semantic atoms and rules

To make the model behavior concrete, we qualitatively inspect individual recommendations and the explanations generated by the semantic horn-rule reranker. Each case contains a user, the top ranked items, and for each item a compact explanation consisting of the routing weight α , the most influential semantic atoms, and the most activated rules. For additional transparency, we further inspect the slot-level matches inside activated rules by tracing which item atoms receive the highest attention and user-atom match. While Table 1 summarizes aggregate performance, these cases illustrate why the same architecture is particularly useful under heterogeneous user histories: the routing mechanism modulates how much the system trusts collaborative evidence versus semantic reasoning, and explanations are produced from the same atoms and rules that contribute to the final score.

On Video_Games, we observe users for whom the learned routing weight is relatively small ($\alpha \approx 0.06$), indicating that the semantic pathway is effectively down-weighted and the model behaves similarly to a pure collaborative scorer for those instances. For one representative user, the top recommendation is an action-adventure game. The explanation highlights semantic atoms such as *open-world exploration*, *multiplayer mode*, together with a small set of consistently activated rules (IDs 12, 7, 22, 3, 18). Because these atoms are exactly the features consumed by the reranker when producing s_{rule} , the model can still offer a coherent rationale even when the semantic contribution to the final score is small.

A complementary diagnostic case arises for users where the learned gate assigns $\alpha = 0$, yielding empty atom lists and no rule activations in the explanation. In such cases, the audit interface explicitly exposes that the recommendation is entirely collaborative in origin. Rather than producing a fabricated semantic rationale, the system surfaces the absence of semantic evidence—which is itself diagnostically informative: it allows practitioners to identify which recommendations are CF-only versus semantically grounded, and to distinguish model failures caused by missing atoms from those caused by incorrect rule activations.

On Yelp2018, the routing weight saturates, indicating a stronger reliance on semantic evidence ($\alpha \approx 1.00$), reflecting the need to rely on item semantics in a high-sparsity environment. For a representative user (ID 19546), the top ranked recommendation (item 10918) is explained by semantic atoms such as restaurant or food service, family-friendly establishment, located in a major city, moderate to high price point, and wi-fi for customers, with activated rules (IDs 7, 24, 30, 25, 10). To make the rule structure concrete, we trace slot-level matches inside activated rules: for example, rule 7 attends to meal-specific suitability and catering services available across multiple slots (slots 1–4), while rule 24 similarly focuses on catering services available (slots 2 and 4) and meal-specific suitability (slot 3). This

illustrates how the reranker aggregates a small set of atom-level signals through a compact, reusable rule structure.

In Anime, we frequently observe intermediate routing weights ($\alpha \approx 0.06$), suggesting that collaborative evidence and semantic reasoning are both informative. For one user (ID 1294), the top recommendation (item 69) is explained by a concise semantic atom such as human world. This provides an actionable summary of the content-level factor the model relied on, while the corresponding rule activations (IDs 28, 31, 17, 15, 1) serve as an explicit trace of the reasoning process.

Across these cases, the explanations remain short but structured, and the routing weight provides a transparent indicator of which pathway dominated the decision. This is important in practice, where cold-start users cannot be treated as an edge case; instead, the system should expose when it is relying on semantic knowledge and how that knowledge affects ranking. In the next section, we further quantify this connection through perturbation-based sensitivity diagnostics that remove semantic atoms and measure the resulting change in scores.

6.5. Mechanism-linked sensitivity analysis via semantic-atom perturbations

A common criticism of explainable recommendation is that explanations can be “plausible but not faithful”, i.e., the produced rationale does not meaningfully reflect the decision mechanism. In our framework, semantic atoms are not an auxiliary annotation; they are part of the scoring computation in the horn-rule reranker. We therefore distinguish two targets of analysis. Pathway-level sensitivity asks whether highlighted atoms materially affect the semantic score inside the reranker. Ranking-level sensitivity asks whether those perturbations are large enough to alter the final ranking after collaborative and routed components are taken into account. We operationalize this idea through counterfactual atom removal at $K = 10$ on 200 sampled users per dataset. For each user-item explanation, we remove either the top- r atoms that appear in the explanation (Top- r) or r randomly chosen atoms (Random- r), and then re-evaluate NDCG@10 under the same candidate set.

We view this protocol as mechanism-linked auditing: it reveals when semantic atoms have an observable marginal contribution to ranking, rather than guaranteeing large ranking changes across all domains and user groups.

To make this diagnostic more informative, we report two complementary faithfulness-style metrics. Deletion (comprehensiveness) compares the ranking change after removing Top- r versus Random- r atoms:

$$\Delta_{del} = \text{NDCG}_{rnd\ rm}(r) - \text{NDCG}_{top\ rm}(r).$$

Sufficiency compares how much of the ranking signal is retained when keeping only Top- r atoms versus only Random- r atoms:

$$\Delta_{suf} = \text{NDCG}_{top\ keep}(r) - \text{NDCG}_{rnd\ keep}(r).$$

We report paired user-level differences with 95% bootstrap confidence intervals and Cohen’s d .

Table 7 shows strong sufficiency effects on Yelp2018 (e.g., $\Delta_{suf} \approx 0.083$ at $r \in \{3, 5\}$ with positive CIs and medium effect sizes), suggesting that the atoms highlighted by the explanation are substantially more informative than random atoms under a strict “only-these-atoms” test. Deletion exhibits a non-trivial separation as well, and the sign for larger budgets indicates redundancy: removing the top-matched atoms can often be compensated by remaining atoms, while removing r random atoms is more likely to disrupt multiple rule activations.

On Video_Games and Anime, both deletion and sufficiency deltas are smaller but detectable under this setup. This attenuated sensitivity is consistent with the model’s routing behavior in these datasets: the routing weight is moderate and the semantic term contributes less than on Yelp2018, so atom-level perturbations produce weaker but measurable influence on the final ranking. On Video_Games, deletion shows mixed signs (negative at $r=5, 10$) indicating that top-atom removal can sometimes improve ranking due to atom redundancy, while sufficiency remains consistently positive, confirming that explanation-highlighted atoms retain more predictive signal than random ones. This remains true even when conditioning on users with higher semantic contribution (top-20% by per-user mean rule-term ratio).

Notably, in this regime ranking-level deltas can vanish even when the semantic term is measurably more sensitive to top-atom perturbations than random ones: Appendix Table 8 reports consistent non-zero deltas in the semantic term magnitude $|\alpha \lambda s_{rule}|$ on the same high-contribution subset, indicating that the highlighted atoms are more influential within the semantic pathway even when the overall pathway magnitude is insufficient to change item ordering.

In contrast, on Yelp2018 the routing weight saturates (mean $\alpha \approx 1.0$) and the semantic term dominates (mean rule-term ratio ≈ 2.18), which is consistent with the clear sufficiency gaps observed in Table 7.

To further characterize the regime dependence of semantic influence at the decision level, we report how often atom perturbations change the final top-10 candidate list. On Yelp2018, removing the top-5 matched atoms alters the top-10 list for 41.0% of users overall; on the top-20% of users by rule-term ratio, this rate rises to 62.5%. Score-level sensitivity confirms the same pattern: keeping top-5 atoms versus random-5 atoms yields a semantic-term sufficiency delta of +1.69 in $|\alpha \lambda s_{rule}|$ for this high-contribution subset, confirming that highlighted atoms materially dominate the semantic pathway even before perturbations propagate to ranking. On Video_Games and Anime, top-10 list changes remain below 15% even in the high rule-ratio subset at $r = 5$, and score-level deltas are near zero, consistent with the weaker gate values in those datasets. Across datasets, these statistics reinforce the conditional nature of ranking-level faithfulness: it is not uniformly strong, but it is strongest precisely where the semantic pathway is most active and mechanically coupled to the final score.

7. Limitations

Several limitations should be acknowledged. First, our perturbation-based evaluation provides mechanism-linked evidence of semantic influence, but the impact is strongly regime-dependent. On Yelp2018, where the routing weight saturates ($\alpha \approx 1.0$) and the semantic term dominates, both pathway-level and ranking-level sensitivity are clearly visible. On Video_Games and Anime, ranking-level faithfulness remains weak: the semantic term contributes little relative to the collaborative score, so atom perturbations rarely propagate to ranking changes. The semantic audit interface retains diagnostic value in these regimes—it explicitly surfaces when semantic reasoning is absent and provides mechanism traces for error analysis—but claims about ranking-level semantic faithfulness should be understood as conditional on the semantic pathway being sufficiently active. A more complete evaluation would include systematic user-level coverage metrics, varying perturbation budgets and atom-list lengths, and ideally an operational human verification across all three regimes.

Second, the three-stage routing shows the largest gains on Video_Games, where a substantial fraction of test users have very sparse histories (18.7% super-cold users with ≤ 2 interactions). On Anime and Yelp2018, the super-cold routing branch is activated less frequently (1.2% and 2.8% of evaluation users, respectively). On Anime the fraction is too small to produce any aggregate improvement (Ours3Stage $\equiv$ Ours); on Yelp2018 the gain is small but measurable (NDCG@10: +0.007). This means that the most visible empirical gains come from a simple and transparent regime policy rather than from a more complex semantic scoring function. We view this as a systems-design choice—favouring explicit and auditable regime handling over a monolithic black-box scorer—but it also narrows the algorithmic novelty claim that can be made for the ranking component alone.

Third, we have added a heuristic routing baseline (HeuristicRoute: cold $\rightarrow$ MostPop, warm $\rightarrow$ LightGCN) in Table 4. Ours3Stage outperforms HeuristicRoute on all three datasets in overall NDCG, confirming the value of semantic routing across regimes. On Video_Games the margin is small (0.537 vs 0.530) because MostPop is unusually effective on that long-tail dataset. This shows that the incremental value of our routing is dataset-dependent but consistently positive.

Fourth, our semantic atoms are extracted from item descriptions using a large language model, which introduces dependency on external knowledge sources and may not generalize to domains without rich textual metadata. Three additional caveats apply. First, LLM API outputs are non-deterministic and may change across versions, so we distinguish rerun stability from experimental reproducibility. To quantify short-term stability, we conducted a rerun experiment on 100 randomly sampled items per dataset, extracting atoms 3 times each with deterministic decoding (temperature 0). Pairwise Jaccard overlap across reruns is 0.82 ± 0.27 (Video_Games) and 0.86 ± 0.22 (Anime), confirming high extraction reproducibility under fixed model

Table 7Mechanism-linked sensitivity analysis via semantic-atom perturbations ($K = 10, 200$ users). Format: mean [95% CI] (Cohen's d).

Dataset	Base NDCG	Deletion Δ_{del}				Sufficiency Δ_{suf}			
		r=1	r=3	r=5	r=10	r=1	r=3	r=5	r=10
VG	0.443	-0.006	0.001	-0.018	-0.014	0.003	0.008	0.012	0.018
		[-0.012,0.001] (-0.05)	[-0.006,0.008] (0.01)	[-0.028,-0.008] (-0.15)	[-0.025,-0.003] (-0.10)	[-0.002,0.008] (0.04)	[0.002,0.014] (0.09)	[0.005,0.019] (0.13)	[0.009,0.027] (0.18)
Anime	0.817	0.001	0.003	0.006	0.010	0.001	0.004	0.007	0.012
		[-0.001,0.003] (0.02)	[0.000,0.006] (0.05)	[0.002,0.010] (0.08)	[0.005,0.015] (0.12)	[-0.001,0.003] (0.02)	[0.000,0.008] (0.06)	[0.003,0.011] (0.09)	[0.006,0.018] (0.14)
Yelp	0.619	0.002	-0.008	-0.014	-0.054	0.056	0.083	0.083	0.040
		[-0.007,0.012] (0.03)	[-0.024,0.008] (-0.08)	[-0.032,0.006] (-0.11)	[-0.076,-0.028] (-0.31)	[0.038,0.077] (0.38)	[0.057,0.109] (0.44)	[0.056,0.109] (0.43)	[0.016,0.066] (0.23)

Note: $\Delta_{del} = \text{NDCG}_{\text{rand rm}}(r) - \text{NDCG}_{\text{top rm}}(r)$; $\Delta_{suf} = \text{NDCG}_{\text{top keep}}(r) - \text{NDCG}_{\text{rand keep}}(r)$.

versions. All downstream experiments use identical frozen JSONL atom artifacts rather than live API calls, ensuring exact reproducibility of the reported results. These artifacts record the source text, extracted atoms, provider/timestamp metadata, and ok/error status for every item, and achieve extraction success rates of >99% on Video_Games, 99.6% on Anime, and 100% on Yelp2018. Second, our manual quality audit reports item-level precision but does not include inter-annotator agreement or atom-level recall/coverage, which limits the strength of the quality evidence. Third, for Yelp2018 we do not compare LLM-derived atoms against direct use of the structured knowledge graph attributes, so whether LLM atomization adds or removes signal relative to raw attributes remains an open question. The quality and coverage of atoms directly affect the semantic reranker's effectiveness.

Fifth, our comparison against neuro-symbolic baselines is limited to DiffFNN as the sole representative. We do not include KGAT [13], RippleNet [16], PGPR [18], or KGCN [19] as direct baselines. These methods require a structured, entity-aligned knowledge graph, whereas our approach uses text-derived semantic atoms; this difference in input assumptions makes direct comparison non-trivial. We acknowledge that including such comparisons would strengthen the empirical conclusions, and we recommend this as a direction for future work. Similarly, we do not include multimodal recommenders (e.g., LATTICE, BM3) or end-to-end LLM-based recommenders (e.g., P5, TALLRec) as baselines. Multimodal methods require aligned image/audio features that are unavailable in our text-only evaluation setting; LLM-based recommenders operate under a fundamentally different inference paradigm (generative rather than score-and-rank) and would require protocol adaptations beyond the scope of this study. Our four added baselines (RLMRec, LightGCL, ELCRec, AlphaRec) cover recent SOTA across LLM-enhanced CF (RLMRec, WWW 2024), graph contrastive learning (LightGCL, ICLR 2023), conversational LLM-enhanced recommendation (ELCRec, NeurIPS 2024), and pluggable LLM-CF paradigms (AlphaRec, ICLR 2025),

providing broad paradigm coverage within the score-and-rank evaluation framework.

Sixth, while we demonstrate statistically significant improvements over DiffFNN, our method does not uniformly dominate all black-box baselines. On Anime, strong collaborative methods (EASE, NeuMF) clearly outperform Ours3Stage in overall NDCG; on Yelp2018, LightGCL and RLMRec are comparable or slightly stronger depending on the metric and K . Only on Video_Games does Ours3Stage outperform all learning-based baselines. Our contribution is positioned as providing a practical accuracy-auditability trade-off rather than state-of-the-art accuracy.

Seventh, our cold-start evaluation focuses primarily on *user* cold-start (users with limited interaction history). We do not present a statistically strong dedicated *new-item* benchmark. Under the current temporal splits, truly unseen test items are rare or absent (Video_Games: 142 unique items, 0.17% of positive test interactions; Anime: 4, 0.01%; Yelp2018: 0), which is why Table 6(b) reports item-popularity bins as a proxy rather than claiming a full item cold-start evaluation. A dataset or split with substantial test-only item coverage would be needed to fully validate item cold-start robustness, and we leave that as future work.

Finally, we do not include a user study to evaluate explanation quality from a human perspective. Such studies would complement our quantitative sensitivity analysis and are left for future work. These limitations do not invalidate the system-level decomposition reported here, but they delimit the scope of the current claims.

8. Conclusion

We presented a regime-aware recommendation architecture that combines shared collaborative and semantic components with history-aware routing. Experiments on three benchmarks (Amazon Video_Games 2023, Anime, Yelp2018) show that routing is the primary source of ranking robustness and the main driver of gains over DiffFNN, while semantic atoms and horn-rule reranking contribute primarily as a knowledge-grounded audit interface rather than as a

universal accuracy booster. On Video_Games, Ours3Stage (NDCG@10=0.537) outperforms all recent learning-based baselines including ELCRec (NeurIPS 2024), AlphaRec (ICLR 2025), RLMRec (WWW 2024), and LightGCL (ICLR 2023), demonstrating competitive accuracy alongside auditable semantic reasoning. The perturbation results further indicate that ranking-level semantic sensitivity is strongest when the semantic pathway materially contributes to the final score, whereas pathway-level sensitivity can remain measurable even when ranking changes are weak. Taken together, these results support a practical accuracy–auditability trade-off and suggest that the main value of the semantic pathway lies in making recommendation decisions inspectable, traceable, and auditable at the knowledge level.

A. Appendix: mechanism-level sensitivity

Table 8: Mechanism-level sensitivity on high-contribution users (top-20%). Format: mean [95% CI] (Cohen's d).

Dataset	r	Deletion Δ_{del}^{score}	Sufficiency Δ_{suf}^{score}
videogames	1	+3.48e-09 [2.87e-09, 4.09e-09] (0.41)	+1.09e-08 [9.87e-09, 1.19e-08] (0.52)
videogames	3	+5.09e-09 [4.42e-09, 5.76e-09] (0.47)	+5.83e-09 [4.97e-09, 6.69e-09] (0.38)
videogames	5	+4.18e-09 [3.66e-09, 4.70e-09] (0.44)	+2.13e-09 [1.56e-09, 2.70e-09] (0.33)
videogames	10	+1.04e-09 [0.91e-09, 1.17e-09] (0.39)	+3.18e-10 [1.78e-10, 4.58e-10] (0.22)
anime	1	+7.03e-08 [6.69e-08, 7.38e-08] (2.76)	+2.05e-07 [1.95e-07, 2.15e-07] (2.80)
anime	3	+8.07e-08 [7.64e-08, 8.54e-08] (2.53)	+8.32e-08 [7.89e-08, 8.77e-08] (2.58)
anime	5	+5.99e-08 [5.63e-08, 6.39e-08] (2.16)	+4.02e-08 [3.79e-08, 4.27e-08] (2.28)
anime	10	+2.51e-08 [2.31e-08, 2.70e-08] (1.75)	+8.20e-09 [7.51e-09, 8.89e-09] (1.59)
yelp2018	1	+0.52 [0.38, 0.66] (1.83)	+0.89 [0.71, 1.07] (2.05)
yelp2018	3	+1.38 [1.15, 1.61] (2.24)	+1.76 [1.52, 2.00] (2.52)
yelp2018	5	+1.51 [1.28, 1.74] (2.41)	+1.69 [1.43, 1.95] (2.38)
yelp2018	10	+1.13 [0.91, 1.35] (1.97)	+1.21 [0.98, 1.44] (2.12)

Note: $\Delta_{del}^{score} = \text{mean}(| \cdot |)_{rnd\ rm}(r) - \text{mean}(| \cdot |)_{top\ rm}(r)$; $\Delta_{suf}^{score} = \text{mean}(| \cdot |)_{top\ keep}(r) - \text{mean}(| \cdot |)_{rnd\ keep}(r)$. Score-level magnitudes are not comparable across datasets because semantic scaling parameters (λ , gate values) are dataset-specific; comparisons should be interpreted within each dataset.

B. Appendix: slot-level rule trace example

Table 9: Slot-level rule trace for Yelp2018 case study (user 19546, item 10918).

Rule	Slot1	Slot2	Slot3	Slot4
7	meal-specific suitability	catering services available	meal-specific suitability	catering services available
24	catering services available	catering services available	meal-specific suitability	catering services available
30	meal-specific suitability	catering services available	catering services available	meal-specific suitability
25	catering services available	meal-specific suitability	meal-specific suitability	catering services available
10	catering services available	meal-specific suitability	catering services available	catering services available

References

- [1] Schein, A. I., Popescul, A., Ungar, L. H., and Pennock, D. M. (2002). Methods and metrics for cold-start recommendations. In *Proceedings of the 25th Annual International ACM SIGIR Conference on Research and Development in Information Retrieval*, pages 253–260.
- [2] Vartak, M., Thiagarajan, A., Miranda, C., Bratman, J., and Larochelle, H. (2017). A meta-learning perspective on cold-start recommendations for items. In *Advances in Neural Information Processing Systems*, pages 6904–6914.
- [3] Volkovs, M., Yu, G., and Poutanen, T. (2017). DropoutNet: Addressing cold start in recommender systems. In *Advances in Neural Information Processing Systems*, pages 4957–4966.
- [4] Hu, Y., Koren, Y., and Volinsky, C. (2008). Collaborative filtering for implicit feedback datasets. In *Proceedings of the 2008 IEEE International Conference on Data Mining (ICDM)*, pages 263–272. DOI: 10.1109/ICDM.2008.22.
- [5] Pan, F., Li, S., Ao, X., Tang, P., and He, Q. (2019). Warm up cold-start advertisements: Improving CTR predictions via learning to learn ID embeddings. In *Proceedings of the 42nd International ACM SIGIR Conference on Research and Development in Information Retrieval*, pages 695–704. DOI: 10.1145/3331184.3331268.
- [6] Adomavicius, G. and Tuzhilin, A. (2005). Toward the next generation of recommender systems: A survey of the state-of-the-art and possible extensions. *IEEE Transactions on Knowledge and Data Engineering*, 17(6):734–749. DOI: 10.1109/TKDE.2005.99.
- [7] Bobadilla, J., Ortega, F., Hernando, A., and Gutiérrez, A. (2013). Recommender systems survey. *Knowledge-Based Systems*, 46:109–132. DOI: 10.1016/j.knsys.2013.03.012.
- [8] Zhang, Y., Chen, X., et al. (2020). Explainable recommendation: A survey and new perspectives. *Foundations and Trends in Information Retrieval*, 14(1):1–101.
- [9] Jacovi, A. and Goldberg, Y. (2020). Towards faithfully interpretable NLP systems: How should we define and evaluate faithfulness? In *Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics*, pages 4198–4205.
- [10] Ribeiro, M. T., Singh, S., and Guestrin, C. (2016). “Why should I trust you?”: Explaining the predictions of any classifier. In *Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery & Data Mining*, pages 1135–1144. DOI: 10.1145/2939672.2939778.
- [11] Zhang, Y., Lai, G., Zhang, M., Zhang, Y., Liu, Y., and Ma, S. (2014). Explicit factor models for explainability in recommender systems. In *Proceedings of the 37th International ACM SIGIR Conference on Research and Development in Information Retrieval*, pages 83–92. DOI: 10.1145/2600428.2609579.
- [12] Wang, X., Wang, D., Xu, C., He, X., Cao, Y., and Chua, T.-S. (2019). Explainable reasoning over knowledge graphs for recommendation. In *Proceedings of the 33rd AAAI Conference on Artificial Intelligence*, pages 5329–5336. DOI: 10.1609/aaai.v33i01.33015329.
- [13] Wang, X., He, X., Cao, Y., Liu, M., and Chua, T.-S. (2019). KGAT: Knowledge graph attention network for recommendation. In *Proceedings of the 25th ACM SIGKDD International Conference on Knowledge Discovery & Data Mining*, pages 950–958.
- [14] Ma, W., Zhang, M., Cao, Y., Jin, W., Wang, C., Liu, Y., Ma, S., and Ren, X. (2019). Jointly learning explainable rules for recommendation with knowledge graph. In *The World Wide Web Conference*, pages 1210–1221.
- [15] Bartl, S., Innerebner, K., and Lex, E. (2025). Differentiable Fuzzy Neural Networks for Recommender Systems. In *Adjunct Proceedings of the 33rd ACM Conference on User Modeling, Adaptation and Personalization (UMAP Adjunct ’25)*.
- [16] Wang, H., Zhang, F., Wang, J., Zhao, M., Li, W., Xie, X., and Guo, M. (2018). RippleNet: Propagating user preferences on the knowledge graph for recommender systems. In *Proceedings of the 27th ACM International Conference on Information and Knowledge Management (CIKM)*, pages 417–426. DOI: 10.1145/3269206.3271739.
- [17] Cao, Y., Wang, X., He, X., Hu, Z., and Chua, T.-S. (2019). Unifying knowledge graph learning and recommendation: Towards a better understanding of user preferences. In *Proceedings of the 2019 World Wide Web Conference (WWW)*, pages 151–161. DOI: 10.1145/3308558.3313705.
- [18] Xian, Y., Fu, Z., Muthukrishnan, S., de Melo, G., and Zhang, Y. (2019). Reinforcement knowledge graph reasoning for explainable recommendation. In *Proceedings of the 42nd International ACM SIGIR Conference on Research and Development in Information Retrieval*, pages 285–294. DOI: 10.1145/3331184.3331203.
- [19] Wang, H., Zhao, M., Xie, X., Li, W., and Guo, M. (2019). Knowledge graph convolutional networks for recommender systems. In *Proceedings of the 2019 World Wide Web Conference (WWW)*, pages 3307–3313. DOI: 10.1145/3308558.3313417.
- [20] Sarwar, B., Karypis, G., Konstan, J., and Riedl, J. (2001). Item-based collaborative filtering recommendation algorithms. In *Proceedings of the 10th International Conference on World Wide Web (WWW)*, pages 285–295. DOI: 10.1145/371920.372071.
- [21] Koren, Y., Bell, R., and Volinsky, C. (2009). Matrix factorization techniques for recommender systems. *Computer*, 42(8):30–37. DOI: 10.1109/MC.2009.263.
- [22] Koren, Y. (2008). Factorization meets the neighborhood: A multifaceted collaborative filtering model. In *Proceedings of the 14th ACM SIGKDD International Conference on Knowledge Discovery & Data Mining*, pages 426–434. DOI: 10.1145/1401890.1401944.
- [23] Rendle, S. (2010). Factorization machines. In *Proceedings of the 2010 IEEE International Conference on Data Mining (ICDM)*, pages 995–1000. DOI: 10.1109/ICDM.2010.127.
- [24] He, X., Zhang, H., Kan, M.-Y., and Chua, T.-S. (2016). Fast matrix factorization for online recommendation with implicit feedback. In *Proceedings of the 39th International ACM SIGIR Conference on Research and Development in Information Retrieval*, pages 549–558. DOI: 10.1145/2911451.2911489.
- [25] Ning, X. and Karypis, G. (2011). SLIM: Sparse linear methods for top- n recommender systems. In *Proceedings of the 2011 IEEE International Conference on Data Mining (ICDM)*, pages 497–506. DOI: 10.1109/ICDM.2011.134.
- [26] Liang, D., Krishnan, R. G., Hoffman, M. D., and Jebara, T. (2018). Variational autoencoders for collaborative filtering. In *Proceedings of the 2018 World Wide Web Conference (WWW)*, pages 689–698. DOI: 10.1145/3178876.3186150.
- [27] Ying, R., He, R., Chen, K., Eksombatchai, P., Hamilton, W. L., and Leskovec, J. (2018). Graph convolutional neural networks for web-scale recommender systems. In *Proceedings of the 24th ACM SIGKDD International Conference on Knowledge Discovery & Data Mining*, pages 974–983. DOI: 10.1145/3219819.3219890.
- [28] Cheng, H.-T., Koc, L., Harmsen, J., Shaked, T., Chandra, T., Aradhye, H., Anderson, G., Corrado, G., Chai, W., Ispir, M., Anil, R., Haque, Z., Hong, L., Jain, V., Liu, X., and Shah, H. (2016). Wide & deep learning for recommender systems. In *Proceedings of the 1st Workshop on Deep Learning for Recommender Systems*, pages 7–10. DOI: 10.1145/2988450.2988454.
- [29] He, X. and Chua, T.-S. (2017). Neural factorization machines for sparse predictive analytics. In *Proceedings of the 40th International ACM SIGIR Conference on Research and Development in Information Retrieval*, pages 355–364. DOI: 10.1145/3077136.3080777.
- [30] Sun, F., Liu, J., Wu, J., Pei, C., Lin, X., Ou, W., and Jiang, P. (2019). BERT4Rec: Sequential recommendation with bidirectional encoder representations from transformer. In *Proceedings of the 28th ACM International Conference on Information and Knowledge Management (CIKM)*, pages 1441–1450. DOI: 10.1145/3357384.3357895.
- [31] Tang, J. and Wang, K. (2018). Personalized top- n sequential recommendation via convolutional sequence embedding. In *Proceedings of the 11th ACM International Conference on Web Search and Data Mining (WSDM)*, pages 565–573. DOI: 10.1145/3159652.3159656.
- [32] Zhang, S., Yao, L., Sun, A., and Tay, Y. (2019). Deep learning based recommender system: A survey and new perspectives. *ACM Computing Surveys*, 52(1):5. DOI: 10.1145/3285029.
- [33] Guo, Q., Zhuang, F., Qin, C., Zhu, H., Xie, X., Xiong, H., and He, Q. (2020). A survey on knowledge graph-based recommender

- systems. *IEEE Transactions on Knowledge and Data Engineering*, 34(8):3549–3568. DOI: 10.1109/TKDE.2020.3028705.
- [34] Ferrari Dacrema, M., Cremonesi, P., and Jannach, D. (2019). Are we really making much progress? A worrying analysis of recent neural recommendation approaches. In *Proceedings of the 13th ACM Conference on Recommender Systems (RecSys)*, pages 101–109. DOI: 10.1145/3298689.3347058.
- [35] Krichene, W. and Rendle, S. (2020). On sampled metrics for item recommendation. In *Proceedings of the 26th ACM SIGKDD International Conference on Knowledge Discovery & Data Mining*, pages 1748–1757. DOI: 10.1145/3394486.3403226.
- [36] Zangerle, E. and Bauer, C. (2022). Evaluating recommender systems: Survey and framework. *ACM Computing Surveys*, 55(8):170. DOI: 10.1145/3556536.
- [37] He, X., Deng, K., Wang, X., Li, Y., Zhang, Y., and Wang, M. (2020). LightGCN: Simplifying and powering graph convolution network for recommendation. In *Proceedings of the 43rd International ACM SIGIR Conference on Research and Development in Information Retrieval*, pages 639–648.
- [38] Rendle, S., Freudenthaler, C., Gantner, Z., and Schmidt-Thieme, L. (2009). BPR: Bayesian personalized ranking from implicit feedback. In *Proceedings of the Twenty-Fifth Conference on Uncertainty in Artificial Intelligence*, pages 452–461.
- [39] Kingma, D. P. and Ba, J. (2015). Adam: A method for stochastic optimization. In *International Conference on Learning Representations (ICLR)*.
- [40] Cremonesi, P., Koren, Y., and Turrin, R. (2010). Performance of recommender algorithms on top-n recommendation tasks. In *Proceedings of the Fourth ACM Conference on Recommender Systems*, pages 39–46.
- [41] He, X., Liao, L., Zhang, H., Nie, L., Hu, X., and Chua, T.-S. (2017). Neural collaborative filtering. In *Proceedings of the 26th International Conference on World Wide Web*, pages 173–182.
- [42] Steck, H. (2019). Embarrassingly shallow autoencoders for sparse data. In *The World Wide Web Conference*, pages 3251–3257.
- [43] Ren, X., Wei, W., Xia, L., Su, L., Cheng, S., Wang, J., Yin, D., and Huang, C. (2024). Representation learning with large language models for recommendation. In *Proc. The Web Conf. (WWW)*, pages 3464–3475.
- [44] Cai, X., Huang, C., Xia, L., and Ren, X. (2023). LightGCL: Simple yet effective graph contrastive learning for recommendation. In *Proc. 11th Int. Conf. on Learning Representations (ICLR)*.
- [45] Wang, C., Yu, Y., Ma, W., Zhang, M., Chen, C., Liu, Y., and Ma, S. (2022). Towards representation alignment and uniformity in collaborative filtering. In *Proc. 28th ACM SIGKDD Conf.*, pages 1816–1825.
- [46] Kang, W.-C. and McAuley, J. (2018). Self-attentive sequential recommendation. In *Proc. IEEE Int. Conf. on Data Mining (ICDM)*, pages 197–206.
- [47] Jacobs, R. A., Jordan, M. I., Nowlan, S. J., and Hinton, G. E. (1991). Adaptive mixtures of local experts. *Neural Computation*, 3(1):79–87.
- [48] Shazeer, N., Mirhoseini, A., Maziarz, K., Davis, A., Le, Q., Hinton, G., and Dean, J. (2017). Outrageously large neural networks: The sparsely-gated mixture-of-experts layer. In *Proc. 5th Int. Conf. on Learning Representations (ICLR)*.
- [49] Du, Z., Xu, L., Zhao, X., Chen, J., Li, Z., Ge, Y., and Zhang, Y. (2024). ELCRec: Enhance language understanding via large language model with contrastive learning for conversational recommendation. In *Advances in Neural Information Processing Systems 37 (NeurIPS)*.
- [50] Hsu, S.-C., Lian, J., Chen, X., and Xie, X. (2025). AlphaRec: Language model empowers collaborative filtering with pluggable paradigm. In *Proc. 13th Int. Conf. on Learning Representations (ICLR)*. (Oral)
- [51] Hou, Y., Li, J., He, Z., Yan, A., Chen, X., and McAuley, J. (2024). Bridging language and items for retrieval and recommendation. In *Proc. 2024 Conf. of the North American Chapter of the Association for Computational Linguistics (NAACL)*.
- [52] Wu, L., Zheng, Z., Qiu, Z., Wang, H., Gu, H., Shen, T., Qin, C., Zhu, C., Zhu, H., Liu, Q., Xiong, H., and Chen, E. (2024). A survey on large language models for recommendation. *World Wide Web*, 27(5):60.
- [53] Wei, W., Ren, X., Tang, J., Wang, Q., Su, L., Cheng, S., Wang, J., Yin, D., and Huang, C. (2024). LLMRec: Large language models with graph augmentation for recommendation. In *Proc. 17th ACM Int. Conf. on Web Search and Data Mining (WSDM)*.
- [54] Bao, K., Zhang, J., Zhang, Y., Wang, W., Feng, F., and He, X. (2023). TALLRec: An effective and efficient tuning framework to align large language model with recommendation. In *Proc. 17th ACM Conf. on Recommender Systems (RecSys)*.
- [55] Li, J., Ren, H., Deng, C., Feng, Y., Mao, K., Zhang, W., and Yu, Y. (2025). CALRec: Contrastive alignment of generative LLMs for sequential recommendation. In *Proc. 13th Int. Conf. on Learning Representations (ICLR)*.
- [56] Xi, Y., Liu, W., Lin, J., Zhu, J., Chen, B., Tang, R., Zhang, W., Zhang, R., and Yu, Y. (2024). Towards open-world recommendation with knowledge augmentation from large language models. In *Proc. 18th ACM Conf. on Recommender Systems (RecSys)*.
- [57] Zheng, B., Hou, Y., Lu, H., Chen, Z., Zhao, W. X., and Wen, J.-R. (2024). Adapting large language models by integrating collaborative semantics for recommendation. In *Proc. IEEE 40th Int. Conf. on Data Engineering (ICDE)*.
- [58] Jia, L., Chen, H., Li, Z., and others (2026). MoToRec: Sparse-regularized multimodal tokenization for cold-start recommendation. In *Proc. 40th AAAI Conf. on Artificial Intelligence (AAAI)*.
- [59] Chen, H., and others (2026). M2VAE: Multi-modal multi-view variational autoencoder for cold-start item recommendation. In *Proc. 40th AAAI Conf. on Artificial Intelligence (AAAI)*.
- [60] Li, H., and others (2025). DisCo: Graph-based disentangled contrastive learning for cold-start cross-domain recommendation. In *Proc. 39th AAAI Conf. on Artificial Intelligence (AAAI)*, pages 33312.
- [61] Huang, F., Bei, Y., Yang, Z., Jiang, J., Chen, H., Shen, Q., Wang, S., Karray, F., and Yu, P. S. (2025). Large language model simulator for cold-start recommendation. In *Proc. 18th ACM Int. Conf. on Web Search and Data Mining (WSDM)*, pages 345–354.
- [62] Kim, J., Kim, H., Cho, H., Kang, S., Chang, B., Yeo, J., and Lee, D. (2025). Review-driven personalized preference reasoning with large language models for explainable recommendation. In *Proc. 48th Int. ACM SIGIR Conf. on Research and Development in Information Retrieval*.